

Weight of Polynomial Products Mod $(X^n + 1)$ –Application to the HQC Cryptosystem–

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Abstract. We consider the following problem: given two random polynomials x and y in the ring $\mathbb{F}_2[X]/(X^n + 1)$, our goal is to compute the expectation and variance of the weight of their product $x \cdot y$, where the weight of a binary polynomial is defined as the number of its nonzero coefficients. We consider two models for random polynomials x and y : the *uniform slice case* with fixed weights w_x, w_y , and the *binomial case* where their coefficients are independent Bernoulli variables with success probabilities p_x and p_y respectively.

Our work finds a direct application in the accurate analysis of the decryption failure rate for the code-based encryption scheme HQC [3]. The original construction [3] relied on heuristic arguments supported by experimental data. Later, [4] provided a formally proven security bound, albeit a much weaker one than the heuristic estimate in [3]. A fundamental limitation of both analyses is their restriction to the binomial case, a simplification that compromises the resulting security guarantees.

Our analysis provides the first precise computation of the expectation and variance of $\text{weight}(x \cdot y)$ across both the uniform slice and binomial models. The results confirm the soundness of the HQC security guarantees and allow for a more informed choice of the scheme parameters that optimizes the trade-off security and efficiency.

Keywords: Weight of a polynomial · Expectation/variance of random variables · Tail probability · Decryption error of HQC

1 Introduction

We consider the following problem. Given a positive integer n , and two polynomials x, y in the ring $\mathbb{F}_2[X]/(X^n + 1)$, we are interested in the Hamming weight – number of nonzero coefficients – of their product. More specifically, we want to compute the expectation and variance of the weight of $x \cdot y \bmod (X^n + 1)$ in the following two models:

1. **The Uniform Slice Model.** The polynomials x and y have fixed weights w_x and w_y respectively.
2. **The Binomial Model.** x, y follow a binomial distribution, i.e., the coefficients of x and y are independent and identically distributed (iid) random variables, each equal to 1 with probability p_x (resp. p_y).

1.1 Motivation

The considered problem finds a direct application in the HQC (Hamming Quasi-Cyclic) [3] code-based encryption scheme, selected by the National Institute of Standards and Technology (NIST) as a backup to the winner of the NIST competition for the first post-quantum cryptography standard (ML-KEM (Kyber) [1]).

In a nutshell, a ciphertext in the HQC scheme consists of a binary codeword (encoding of a plaintext) to which is added a random binary error vector. Decryption of such a ciphertext is performed by eliminating most of the errors using the decryption key to obtain a noisy version

of the plaintext, then recovering the (noiseless) plaintext using an efficient decoding algorithm. Since the error vector is randomly generated, the scheme HQC has a nonzero decryption failure rate, meaning that decryption occasionally fails with a given probability. In fact, to be able to efficiently recover the plaintext, the weight of the error vector underlying the plaintext must not exceed the error-correction capacity of the underlying code. This requires a careful study of the error vector e' in the HQC scheme which is given by $e' = x \cdot r_2 + y \cdot r_1 + e$, where x, y, r_1, r_2, e are independent uniformly random polynomials in $\mathbb{F}_2[X]/(X^n + 1)$ with fixed weights w, w, w_1, w_2, w_e respectively.

It is clear that the security analysis of the HQC scheme involves studying the expectation and variance of the weight of polynomials product modulo $X^n + 1$.

1.2 Related Work

Let x, y be two independent uniformly random polynomials in the ring $\mathbb{F}_2[X]/(X^n + 1)$ with respective weights w_x and w_y . Let further z denote the product $x \cdot y$ (modulo $X^n + 1$).

The first work to address the expectation and variance of the weight of z was [3], where the authors introduced two key simplifying assumptions:

1. The coefficients of the polynomials x and y are pairwise independent.
2. The coefficients of z itself are pairwise independent.

They were therefore able to derive the expectation and variance of the weight of z , and they supported their heuristics with empirical evidence.

When applied to the HQC scheme, the first assumption is plausible but introduces both computational overhead and a reduction in security guarantees. Specifically, the key generation and encryption algorithms must be modified to produce vectors with pairwise independent coordinates rather than random vectors of a predefined weight, which incurs additional computational cost. Furthermore, the security proof requires revision to accommodate the independence assumption. This adaptation often undermines the proof's tightness, as it necessitates repeating security games until the vectors meet the prescribed weight conditions.

The second assumption—the pairwise independence of z 's coefficients—is however, questionable. Recent work by Bombar et al. [2] provides evidence that this model fails to hold in certain cases. This implies that the product z is a bit more subtle than previously predicted.

Another significant contribution to this problem is the work of Kawachi [4]. The author also adopts the first assumption—independence of the coefficients of the factors x and y —, which facilitates the computation of the expectation. A key contribution of the work is an upper bound on the variance of $\text{weight}(z)$ that is polynomial in the parameter n if the weight of the factors x and y is small enough (around $\sqrt{n}$).

Although this bound is asymptotically weaker than the one in [3], Kawachi's analysis provides a formal theoretical proof in lieu of relying on a heuristic assumption.

A further limitation of the Kawachi analysis is its lack of direct applicability to the HQC error vector. Specifically, a bound on the variance of the weight of $x \cdot y$ does not directly yield a corresponding bound on the variance of the weight of the HQC error vector.

1.3 Our Contributions

We make the following primary contributions:

1. **Explicit Formulae for the expectation and variance of polynomials product.**

we derive exact expressions for the expectation and variance of the weight of the product $z = x \cdot y$ for random polynomials x and y in $\mathbb{F}_2[X]/(X^n + 1)$ under two distinct distributions:

- **The Uniform Slice Model.** We provide formulae for the expectation and variance of the weight of z when x, y are independent random polynomials with fixed weights w_x and w_y . Our analysis demonstrates that the weight of z is sharply concentrated around its mean for almost the entire parameter space (Figure 1a).
- **The Binomial Model.** We provide explicit formulae for the variance of the weight of z when x, y are independent random polynomials with coordinates drawn from Bernoulli distributions with parameters p_x and p_y respectively. In this model, strong concentration of the weight of z is shown to hold only for large Bernoulli parameters (larger than $\frac{1}{\sqrt{n}}$) (Figure 1a).

Furthermore, our analytical study allows us to evaluate existing approximations for the binomial case: the method proposed by [3] is confirmed to be sufficiently accurate for large Bernoulli parameters (Figure 1a). In contrast, the approximation by [4] is demonstrated to give irrelevant upper bounds for the tail probability of the weight of z , as quantitatively illustrated in Figure 1b.

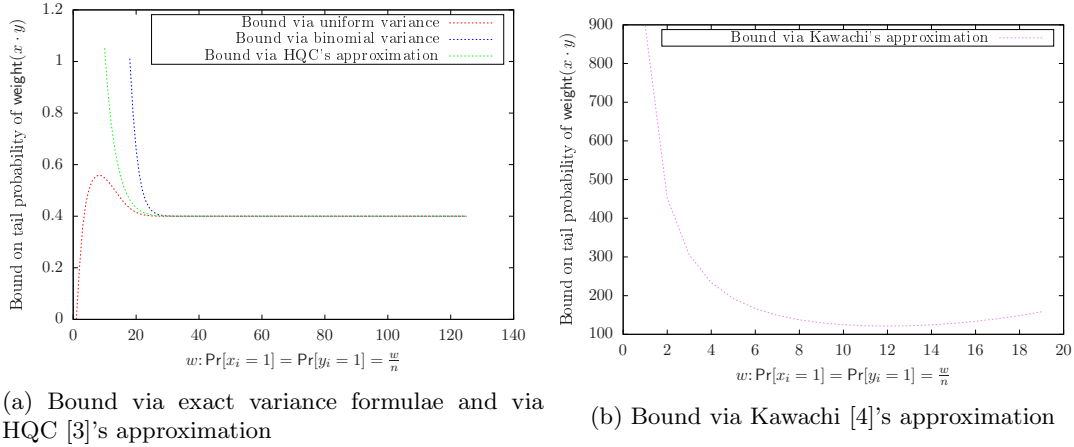


Fig. 1: Upper bound on the tail probability $\Pr[\text{weight}(x \cdot y) \notin (1 \pm 0.1)\mathbb{E}(\text{weight}(x \cdot y))]$ for $n = 250$ and $w \in [0, 125]$

2. Application to the HQC scheme.

We apply our exact formulae to conduct a security reassessment of the HQC scheme. By computing exactly the expectation and variance of the error vector, we conclude that the security guarantees provided in [3] are sound for the uniform slice instantiation, but their validity for the binomial instantiation is limited to scenarios with large Bernoulli parameters. Furthermore, our derived formulae provide a concrete foundation for optimizing system parameters to achieve tighter concentration bounds for the error vector and, consequently, enhanced security guarantees.

2 Overview of Our Techniques

2.1 Tools and Notations

Generalities. $\mathbb{Z}$ denotes the set of integers, and $\mathbb{N}$ denotes the set of natural numbers. The absolute value of an integer i is denoted $|i|$. For real numbers a, b , $[a, b]$ and (a, b) denote respectively the closed and open intervals with bounds a and b . For nonnegative integers n and k , $\binom{n}{k}$ denotes the binomial coefficient defined to be 0 if $k > n$, and $\frac{n!}{k!(n-k)!}$ otherwise.

Let $a, b, n \in \mathbb{Z}$, with $n > 0$. $\mathbb{Z}_n$ denotes the ring of integers modulo n . In this work, the modulo notation is used in two distinct but related ways, discernable from context. First, the expression $b = a \bmod n$ denotes the binary operation yielding the remainder of the Euclidean division of a by n , which is the unique integer b satisfying $0 \leq b < n$. Second, the ternary congruence relation $a \equiv b \pmod{n}$ signifies that a and b are congruent modulo n , which holds if and only if n divides the difference $a - b$. Finally, if $a \equiv b \pmod{n}$ and $0 \leq b \leq n - 1$, then $b = a \bmod n$.

$\mathbb{F}_2[X]/(X^n + 1)$ denotes the ring of binary polynomials modulo the polynomial $X^n + 1$, where addition and multiplication are performed modulo $X^n + 1$. In the text, polynomials in $\mathbb{F}_2[X]/(X^n + 1)$ are identified with their coefficient vector in $\{0, 1\}^n$. For a polynomial (or n -bit vector) $x = (x_0, \dots, x_{n-1})$, $\text{weight}(x)$ denotes the weight, i.e. the number of nonzero coordinates, of x .

Let A and B be two sets. Then the intersection of A and B is denoted $A \cap B = \{x: x \in A, x \in B\}$. The set difference $A \setminus B$ refers to the set $\{x: x \in A, x \notin B\}$. The size of set A is denoted $|A|$.

Finally, we analyze our algorithms using the Big- O notation which gives the worst-case complexity of an algorithm.

Probabilities. The PMF of a random variable denotes its probability mass function. $\mathbb{E}(X)$ and $\text{Var}(X)$ denote respectively the expectation and variance of random variable X . The covariance $\text{Cov}(X_1, X_2)$ of two binary random variable X_1 and X_2 is given by $\Pr[X_1 = 1, X_2 = 1] - \Pr[X_1 = 1]\Pr[X_2 = 1]$. If $X = X_1 + \dots + X_n$ is the sum of n binary random variables, then $\text{Var}(X)$ can be computed as $\sum_{1 \leq k, l \leq n} \text{Cov}(X_k, X_l)$.

Let n be a positive integer, and p a real number in $[0, 1]$. For a binary random variable X , the expression $X \sim \text{Bernoulli}(p)$ signifies that X takes the value 1 with probability p and the value 0 with probability $1 - p$. For binary random variables $X_1, \dots, X_n$, the expression $(X_1, \dots, X_n) \sim \text{Binomial}(n, p)$ means that the X_i are independent and identically distributed (iid) with $X_i \sim \text{Bernoulli}(p)$, for integers $i \in [1, n]$.

Let X be a random variable. The probability that a X deviates by a given amount from its expectation is referred to as a tail probability. **Chebyshev's inequality** gives an upper bound to this probability for any $\delta \in (0, 1)$:

$$\Pr[|X - \mathbb{E}(X)| > \delta \mathbb{E}(X)] < \frac{\text{Var}(X)}{(\delta \mathbb{E}(X))^2}$$

Graphs. A graph G is given by its set of vertices, denoted $V(G)$, and its set of edges, denoted $E(G)$. The edge connecting vertex v_i to vertex v_j is denoted (v_i, v_j) . An undirected graph is a graph where edges represent a two-way bidirectional relationship.

A cycle graph C is an undirected graph that consists of a single cycle; if $V(C) = \{v_1, \dots, v_n\}$, then C is denoted in the text $v_1 - \dots - v_n - v_1$.

A linear chain L is an undirected graph such that all the vertices are connected in a single sequence without cycles; if $V(L) = \{v_1, \dots, v_n\}$, then L is denoted in the text $v_1 - \dots - v_n$.

Dynamic Programming. Our results rely fundamentally on dynamic programming as a central mathematical tool. This approach decomposes the complex problem into simpler sub-problems, each solved exactly once with its solution stored for future reference. Our formulae derived using this method consist of a recurrence relation that defines how to compute each term from the preceding terms, complemented by well-specified base conditions that anchor the computation. We implement this through a bottom-up methodology, where we begin by the base conditions and iteratively progress to compute subsequent terms. Throughout this process, we maintain a

state table that records all intermediate solutions until we ultimately obtain the final solution to our original problem.

2.2 Overview of Our Approach

Let Z be the weight of $x \cdot y$ for two random polynomials $x, y \in \mathbb{F}_2[X]/(X^n + 1)$. Our goal is to compute the expectation and variance of Z . First, we write $Z = Z_0 + \dots + Z_{n-1}$ as a sum of n binary random variables Z_i for $i \in [0, n-1]$.

Since the expectation is linear, then it suffices to compute the expectation of each Z_i . We compute $\mathbb{E}(Z_i)$ by following a recursive approach. We present two main options for the solution. The first is to use dynamic programming, which calculates the result step-by-step in a bottom-up manner. The second is to solve the recursion directly to find a closed formula for $\mathbb{E}(Z_i)$ using combinatorial mathematics.

For the variance, we compute $\text{Var}(Z)$ as the sum of the covariance coefficients $\text{Cov}(Z_k, Z_l)$ for $0 \leq k, l \leq n-1$. It is not hard to see that $\text{Cov}(Z_k, Z_k)$ is nothing but $\mathbb{E}(Z_k)(1 - \mathbb{E}(Z_k))$. The primary challenge resides then in computing $\text{Cov}(Z_k, Z_l)$, and more specifically the joint probability $\Pr[Z_k = 1, Z_l = 1]$, for $k \neq l$. To address this, we introduce an auxiliary random variable, say S , to loosen the correlation between Z_k and Z_l ; $\Pr[Z_k = 1, Z_l = 1]$ is completely determined by the PMF of the introduced random variable. The computation of the PMF of S is a pivotal result of this work as it spans two sections of the paper. Actually, we give a graphic representation to this problem, and reduce computing the PMF to computing the number of edges with prescribed properties in a given graph. Then, we show how to compute this number of edges in case of a cycle graph; we achieve this via auxiliary recurrent functions that we compute in a dynamic programming manner. Then, using basic number theory, we prove that the involved graph in the PMF is either a cycle graph or a disjoint union of cycle graphs. This disjoint property allows independence of the involved probabilities and permits consequently to determine completely the PMF of the introduced random variable.

3 Expectation in the Uniform Slice Model

3.1 Analytical Derivation

Let x, y be independent and uniformly random polynomials from $\mathbb{F}_2[X]/(X^n + 1)$ with $x = (x_0, \dots, x_{n-1})$, $y = (y_0, \dots, y_{n-1})$, $\text{weight}(x) = w_x$ and $\text{weight}(y) = w_y$. Let further $z = (z_0, \dots, z_{n-1})$ denote the product $x \cdot y$. Then $z_k = \bigoplus_{i+j \equiv k \pmod{n}} x_i y_j$. Now, if $Z = \text{weight}(z)$, and $Z_k = \text{weight}(z_k)$. Then by linearity of the expectation, we get $\mathbb{E}(Z) = \sum_{k=0}^{n-1} \mathbb{E}(Z_k)$. On the other hand, and since Z_k is a binary random variable, then $\mathbb{E}(Z_k) = \Pr[Z_k = 1] = \Pr[z_k = 1]$. It suffices then to compute $\Pr[\bigoplus_{i+j \equiv k \pmod{n}} x_i y_j = 1]$ for random binary polynomials x, y . We will proceed in a dynamic programming approach.

First, note that $\Pr[\bigoplus_{i+j \equiv k \pmod{n}} x_i y_j = 1] = \Pr[\bigoplus_{i=0}^{n-1} x_i y_{(k-i) \bmod n} = 1]$. There are consequently n tuples $(x_i, y_{(k-i) \bmod n})$, pairwise disjoint, that are involved in the computation of z_k . Let us denote $t_{i,k} = x_i \cdot y_{(k-i) \bmod n}$. Then, the probability that $t_{i,k}$ equals 1 is $\frac{\binom{w_x-1}{n-1} \binom{w_y-1}{n-1}}{\binom{w_x}{n} \binom{w_y}{n}}$ or equivalently $\frac{w_x \cdot w_y}{n^2}$ since x and y are independent from each other and have weights w_x and w_y respectively.

$$\begin{aligned}
\Pr[z_k = 1] &= \Pr\left[\bigoplus_{i=0}^{n-1} t_{i,k} = 1\right] \\
&= \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1, t_{n-1,k} = 0\right] + \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 0, t_{n-1,k} = 1\right] \\
&= \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1, t_{n-1,k} = 0\right] + \Pr[t_{n-1,k} = 1] \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 0 \mid t_{n-1,k} = 1\right] \\
&= \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1, t_{n-1,k} = 0\right] + \Pr[t_{n-1,k} = 1] (1 - \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1 \mid t_{n-1,k} = 1\right]) \\
&= \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1\right] + \Pr[t_{n-1,k} = 1] - 2 \Pr[t_{n-1,k} = 1] \Pr\left[\bigoplus_{i=0}^{n-2} t_{i,k} = 1 \mid t_{n-1,k} = 1\right]
\end{aligned}$$

Note that $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1]$, for a given N in $[0, n]$, depends solely on N (the number of summands) and $\Pr[t_{i,k} = 1] = \frac{w_x w_y}{n^2}$. More precisely, the probability satisfies the following equation:

$$\Pr\left[\bigoplus_{i=0}^{N-1} t_{i,k} = 1\right] = \Pr[t_{N-1,k} = 1] + \Pr\left[\bigoplus_{i=0}^{N-2} t_{i,k} = 1\right] - 2 \Pr[t_{N-1,k} = 1] \Pr\left[\bigoplus_{i=0}^{N-2} t_{i,k} = 1 \mid t_{N-1,k} = 1\right] \quad (1)$$

Now, let us consider $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{N,k} = 1]$. Again, note that this probability depends only on N (the number of summands) and $\Pr[t_{i,k} = 1 \mid t_{N,k} = 1]$, for $i \in [0, N-1]$, where

$$\begin{aligned}
\Pr[t_{i,k} = 1 \mid t_{N,k} = 1] &= \Pr[t_{i,k} = 1 \mid x_N y_{(k-N) \bmod n} = 1] \\
&= \Pr[x_i = 1, y_{(k-i) \bmod n} = 1 \mid x_N = 1, y_{(k-N) \bmod n} = 1] \\
&= \Pr[x_i \mid x_N = 1] \Pr[y_{(k-i) \bmod n} = 1 \mid y_{(k-N) \bmod n} = 1] \quad (x \text{ and } y \text{ are independent from each other}) \\
&= \frac{(w_x - 1)(w_y - 1)}{(n-1)^2}
\end{aligned}$$

Thus, $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{N,k} = 1]$ depends solely on N and $\frac{(w_x-1)(w_y-1)}{(n-1)^2}$.

More generally, $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{M,k} = 1]$, for $M \notin [0, N-1]$, depends only N and $\frac{(w_x-1)(w_y-1)}{(n-1)^2}$.

If we iterate the same argument, then $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{M_1,k} = 1, \dots, t_{M_j,k} = 1]$, for $1 \leq j \leq \min(w_x, w_y)$ and pairwise distinct $M_j \notin [0, N-1]$, depends solely on N and $\frac{(w_x-j)(w_y-j)}{(n-j)^2}$.

On the other hand, let w denote $\min(w_x, w_y)$. Define further $p_j = \frac{(w_x+j-w)(w_y+j-w)}{(n+j-w)^2}$, for $0 \leq j \leq w$. For instance, we have $p_0 = 0$, $p_{w-1} = \frac{(w_x-1)(w_y-1)}{(n-1)^2}$, and $p_w = \frac{w_x w_y}{n^2}$ and so on. Finally consider the following recursive function.

$$\begin{aligned}
e: ([0, n] \times [0, w]) \cap (\mathbb{Z})^2 &\longrightarrow [0, 1] \\
(i, j) &\longmapsto \begin{cases} 0 & \text{if } \{i = 0 \text{ or } j = 0\} \\ e(i-1, j) + p_j(1 - 2e(i-1, j-1)) & \text{otherwise} \end{cases} \quad (2)
\end{aligned}$$

Theorem 1 (Expectation Formula)

Let $x, y \in \mathbb{F}_2[X]/(X^n + 1)$ two random binary polynomials with respective weights w_x and w_y . Let further $z = x \cdot y$. Then the expectation of $Z = \text{weight}(z)$ is given by $n \cdot e(n, \min(w_x, w_y))$, where e is the function defined in 2.

Proof. From the analysis above, if $z = (z_0, \dots, z_{n-1})$, then $\mathbb{E}(Z) = \sum_{i=0}^{n-1} \mathbb{E}(z_k)$. Also, with the notations above, we have $\mathbb{E}(z_k) = \Pr[z_k = 1] = \Pr[\bigoplus_{i=0}^{n-1} t_{i,k} = 1]$. Since the probability $\Pr[\bigoplus_{i=0}^{n-1} t_{i,k} = 1]$ does not depend on k , then it suffices to prove that it is equal to $e(n, \min(w_x, w_y))$. We do this by induction on n .

Let $w = \min(w_x, w_y)$. First for $n = 0$, i.e., the number of summands is zero, then clearly the probability $\Pr[\bigoplus_{i=0}^{n-1} t_{i,k} = 1]$ is zero, and so is $e(0, w)$ by definition. For $n = 1$, then $\Pr[\bigoplus_{i=0}^{n-1} t_{i,k} = 1]$ boils down to $\Pr[t_{0,k} = 1] = \frac{w_x w_y}{n^2} = p_w$. From the other hand, $e(1, w) = p_w$ according to the recurrence equation of e .

Now suppose the property holds true until N in $[1, n-1]$. Then $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1] = e(N, w)$

According to Equation 1 and the induction hypothesis:

$$\begin{aligned} \Pr[\bigoplus_{i=0}^N t_{i,k} = 1] &= \Pr[t_{N,k} = 1] + \Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1] - 2\Pr[t_{N,k} = 1] \Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{N,k} = 1] \\ &= p_w + e(N, w) - 2p_w e(N, w-1) \\ &= e(N+1, w) \end{aligned}$$

The penultimate equation follows because the probability $\Pr[\bigoplus_{i=0}^{N-1} t_{i,k} = 1 \mid t_{N,k} = 1]$ considers the sum of N terms, where each term has a probability $\frac{(w_x-1)(w_y-1)}{(n-1)^2} = p_{w-1}$ of being one. This probability is then given by $e(N, w-1)$. $\square$

3.2 Practical Computation

We present two main options to compute the expectation. The first is to use dynamic programming, whereas the second consists in solving the recurrence underlying function e defined in Equation 2 in order to obtain a closed formula.

Dynamic Programming. We proceed in bottom-up manner. That is, we first compute $e(i, j)$ for the initial values $i = 0$ and $j = 0$, then use the recurrence equation to compute $e(i, j)$ for larger values. We will need a 2-dimensional table $n \times \min(w_x, w_y)$ for this process. Thus, this approach computes the expectation in time and memory in $O(n \min(w_x, w_y))$.

A Closed Formula for the Expectation. We now solve the recurrence for the function e defined in Equation 2 to obtain a closed formula, thereby also finding a closed formula for the expectation. We first need the following lemma.

Lemma 1. *Let i, s be positive integers with $i > s$. Then*

$$\sum_{t_1=1}^{i-1} \sum_{t_2=1}^{t_1-1} \dots \sum_{t_s=1}^{t_{s-1}-1} t_s = \binom{i}{s+1}$$

Proof (By induction on s). For $s = 1$, we have: $\sum_{t_1=1}^{i-1} t_1 = \frac{i(i-1)}{2} = \binom{i}{2}$.

Suppose $s > 1$, and that the property holds true for $s - 1$. Then

$$\begin{aligned}
\sum_{t_1=1}^{i-1} \sum_{t_2=1}^{t_1-1} \dots \sum_{t_s=1}^{t_{s-1}-1} t_s &= \sum_{t_1=1}^{i-1} \left[\sum_{t_2=1}^{t_1-1} \dots \sum_{t_s=1}^{t_{s-1}-1} t_s \right] \\
&= \sum_{t_1=1}^{i-1} \left[\sum_{r_1=1}^{t_1-1} \dots \sum_{r_{s-1}=1}^{r_{s-2}-1} r_{s-1} \right] \\
&= \sum_{t_1=1}^{i-1} \binom{t_1}{s} \quad (\text{Induction hypothesis for } s-1 \text{ and } i=t_1) \\
&= \binom{s}{s} + \dots + \binom{i-1}{s}
\end{aligned}$$

On the other hand

$$\begin{aligned}
\binom{i}{s+1} &= \binom{i-1}{s+1} + \binom{i-1}{s} \\
&= \binom{i-2}{s+1} + \binom{i-2}{s} + \binom{i-1}{s} \\
&= \binom{i-k}{s+1} + \sum_{t=i-k}^{i-1} \binom{t}{s} \\
&= \binom{s}{s+1} + \sum_{t=s}^{i-1} \binom{t}{s} \quad (k = i - s) \\
&= \binom{s}{s} + \dots + \binom{i-1}{s}
\end{aligned}$$

□

Theorem 2 (Closed Formula for Function e)

The function e defined in Equation 2 satisfies

$$e(i, j) = \sum_{s=0}^{j-1} (-2)^s \frac{\binom{i}{s+1} \binom{w_x-w+j}{s+1} \binom{w_y-w+j}{s+1}}{\binom{n-w+j}{s+1}^2}$$

Proof. The function e satisfies $e(i, j) = e(i-1, j) + p_j(1 - 2e(i-1, j-1))$, with $e(i, 0) = e(0, j) = 0 \quad \forall i, j$. If we expand the recurrence, we get

$$\begin{aligned}
e(i, j) &= p_j + e(i-1, j) - 2p_j e(i-1, j-1) \\
&= p_j + p_j + e(i-2, j) - 2p_j(e(i-1, j-1) + e(i-2, j-1)) \\
&= kp_j + e(i-k, j) - 2p_j \sum_{t_1=1}^k e(i-t_1, j-1) \\
&= (i-1)p_j + e(1, j) - 2p_j \sum_{t_1=1}^{i-1} e(i-t_1, j-1) \quad (k = i-1) \\
&= ip_j - 2p_j \sum_{t_1=1}^{i-1} e(i-t_1, j-1) \\
&= ip_j - 2p_j \sum_{t_1=1}^{i-1} ((i-t_1)p_{j-1} - 2p_{j-1} \sum_{t_2=1}^{i-t_1-1} e(i-t_1-t_2, j-2)) \quad (\text{apply previous equation to } e(i-t_1, j-1)) \\
&= ip_j - 2p_j p_{j-1} \sum_{t_1=1}^{i-1} (i-t_1) + (-2)^2 p_j p_{j-1} \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} e(i-t_1-t_2, j-2) \\
&= ip_j + (-2)p_j p_{j-1} \sum_{t_1=1}^{i-1} (i-t_1) + \dots + (-2)^{s-1} p_j \dots p_{j-s+2} \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_{s-1}=1}^{i-t_1-\dots-t_{s-2}-1} (i-t_1-\dots-t_{s-1}) p_{j-s+1} + \\
&\quad (-2)^s p_j \dots p_{j-s+1} \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_s=1}^{i-t_1-\dots-t_{s-1}-1} e(i-t_1-\dots-t_s, j-s) \\
&= ip_j + (-2) \cdot p_j p_{j-1} \sum_{t_1=1}^{i-1} (i-t_1) + \dots + (-2)^{j-2} p_j \dots p_3 \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_{j-2}=1}^{i-t_1-\dots-t_{j-3}-1} (i-t_1-\dots-t_{j-2}) p_2 + \\
&\quad (-2)^{j-1} p_j \dots p_2 \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_{j-1}=1}^{i-t_1-\dots-t_{j-2}-1} e(i-t_1-\dots-t_{j-1}, 1) \\
&= (-2)^0 \cdot \binom{i}{1} p_j + (-2)^1 \cdot p_j p_{j-1} \sum_{t_1=1}^{i-1} (i-t_1) + \dots + (-2)^{j-2} p_j \dots p_3 p_2 \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_{j-2}=1}^{i-t_1-\dots-t_{j-3}-1} (i-t_1-\dots-t_{j-2}) + \\
&\quad (-2)^{j-1} p_j \dots p_2 p_1 \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_{j-1}=1}^{i-t_1-\dots-t_{j-2}-1} (i-t_1-\dots-t_{j-1})
\end{aligned}$$

The last equation is due to expanding $e(i-t_1-\dots-t_{j-1}, 1)$ using the recurrence to obtain $(i-t_1-\dots-t_{j-1})p_1$ since $e(k, 0) = 0 \quad \forall k$.

Now, using induction or equivalently a series of change of variables, the general term of the sum, namely

$$(-2)^s p_j \dots p_{j-s} \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{i-t_1-1} \dots \sum_{t_s=1}^{i-t_1-\dots-t_{s-1}-1} (i-t_1-\dots-t_s)$$

boils down to

$$(-2)^s p_j \dots p_{j-s} \sum_{t_1=1}^{i-1} \sum_{t_2=1}^{t_1} \dots \sum_{t_s=1}^{t_{s-1}-1} t_s$$

which comes to $(-2)^s p_j \dots p_{j-s} \binom{i}{s+1}$ according to Lemma 1.

On the other hand,

$$\prod_{k=0}^s p_{j-k} = \frac{\binom{w_x-w+j}{s+1} \binom{w_y-w+j}{s+1}}{\binom{n-w+j}{s+1}^2}$$

□

Corollary 1. Let $x, y \in \mathbb{F}_2[X]/(X^n + 1)$ be two random binary polynomials with respective weights w_x and w_y . Let further $z = x \cdot y$. Then the expectation of $Z = \text{weight}(z)$ is given by

$$\mathbb{E}(Z) = n \cdot \sum_{s=0}^{\min(w_x, w_y)-1} (-2)^s \frac{\binom{w_x}{s+1} \binom{w_y}{s+1}}{\binom{n}{s+1}}$$

□

3.3 Validation and Comparative Analysis

We validate the formula for the expectation, computed according to Corollary 1, experimentally. We computed an empirical value for the expectation by averaging over 100,000 trials, where each trial involved generating two random w -weighted polynomials of degree less than $n = 250$ and calculating the weight of their product modulo $X^n + 1$. The close alignment between the curves labeled '**Theory**' and '**Experiments**' in Figure 2a confirms the accuracy of the formula provided in Corollary 1.

On the other hand, we compare in Figure 2b the expectation in the uniform slice and the binomial settings for $n = 250$.

The expectation of two polynomials with Bernoulli($\frac{w}{n}$) coefficients is given by $\frac{n}{2}(1 - (1 - 2(\frac{w}{n})^2)^n)$, according to [3,4] (can also be easily derived from Lemma 2).

The curve labeled '**Uniform distribution**' represents the expectation of the weight of the product modulo $X^n + 1$ of two uniformly random w -weighted polynomials, while the '**Binomial distribution**' curve represents the expectation of the product modulo $X^n + 1$ of two random polynomials with Bernoulli($\frac{w}{n}$) coefficients. Within the interval $[0, 25]$, the uniform expectation slightly exceeds the binomial expectation. The curves converge over the range $[25, 230]$. Beyond this, in the interval $[230, 250]$, the uniform expectation exhibits steady growth, while the binomial expectation remains constant.

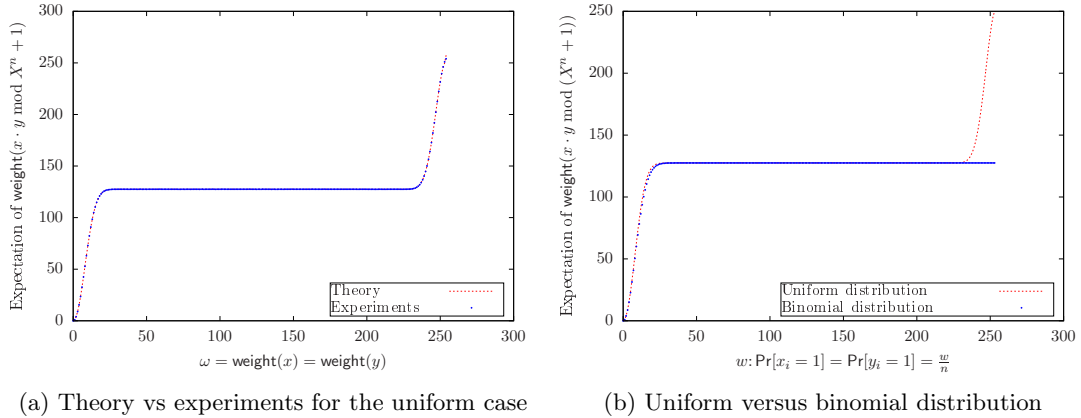


Fig. 2: Expectation of the weight of polynomial products modulo $X^n + 1$

4 Variance

Let x, y be random elements from $\mathbb{F}_2[X]/(X^n + 1)$. Let further $z = (z_0, \dots, z_{n-1})$ denote the product $x \cdot y \bmod (X^n + 1)$. In this section, we compute the variance of $Z = \text{weight}(z)$ across the following models:

1. **The Uniform Slice Model:** x and y are uniform random n -bit vectors with fixed weights w_x and w_y respectively.
2. **The Binomial Model:** x and y are n -bit vectors, with iid Bernoulli(p_x) and Bernoulli(p_y) coordinates respectively.

We know that $Z = \sum_{k=0}^n Z_k$, with $Z_k = \bigoplus_{0 \leq i < n} x_i y_{(k-i) \bmod n}$. Note that the Z_k 's are not independent, thus we cannot apply linearity to compute $\text{Var}(Z)$ as the sum of $\text{Var}(Z_k)$. We use then the standard definition of the variance, namely $\text{Var}(Z) = \sum_{k,l} \text{Cov}(Z_k, Z_l)$ with

$$\text{Cov}(Z_k, Z_l) = \Pr[Z_k = 1, Z_l = 1] - \Pr[Z_k = 1] \Pr[Z_l = 1]$$

It is easy to see that $\text{Cov}(Z_k, Z_k)$ equals $\mathbb{E}(Z_k)(1 - \mathbb{E}(Z_k))$, where $\mathbb{E}(Z_k)$ has been already computed in the previous section. It remains to compute $\text{Cov}(Z_k, Z_l)$, more precisely, the joint probability $\Pr[Z_k = 1, Z_l = 1]$ in case $k \neq l$. Without loss of generality, we assume $k < l$ in the rest of this section.

We drew upon the idea of [4] at the outset. That is, we try to loosen the correlation between Z_k and Z_l by assigning values to x . In this case Z_k and Z_l reduce to $\bigoplus_{i \in S_k} Y_i$ and $\bigoplus_{i \in S_l} Y_i$ respectively, where S_k and S_l denote the sets of surviving indices following the assignment of a value to x . Let us give an example for the uniform slice case. Suppose $n = 5$, $(k, l) = (3, 4)$, and $w_x = 3$. Suppose further that we assign the value $(1, 0, 1, 1, 0)$ to $x = (x_0, x_1, x_2, x_3, x_4)$. Then, under this assignment, $Z_k = \bigoplus_{0 \leq i < n} x_i y_{(k-i) \bmod n}$ becomes $Z_3 = y_0 \oplus y_1 \oplus y_3$ and $Z_l = \bigoplus_{0 \leq i < n} x_i y_{(l-i) \bmod n}$ becomes $Z_4 = y_1 \oplus y_2 \oplus y_4$. In other terms, the surviving indices for the y_i 's involved in Z_k and Z_l are $\{0, 1, 3\}$ and $\{1, 2, 4\}$ respectively. More generally, we define the set of surviving indices for $k \in [0, n-1]$ as follows

$$S_k = \{(k-i) \bmod n : x_i = 1\}$$

One can easily observe that the size of S_k equals w_x regardless of k . Back to the previous example, once x had been assigned the value $(1, 0, 1, 1, 0)$, the joint probability $\Pr[Z_3 = 1, Z_4 = 1, x = (1, 0, 1, 1, 0)]$ becomes

$$\begin{aligned} \Pr[Z_3 = 1, Z_4 = 1, x = (1, 0, 1, 1, 0)] &= \Pr[y_0 \oplus y_1 \oplus y_3 = 1, y_1 \oplus y_2 \oplus y_4 = 1] \\ &= \sum_{b \in \{0,1\}} \Pr[Y_1 = b, y_0 \oplus y_3 = 1 - b, y_2 \oplus y_4 = 1 - b] \end{aligned}$$

And more generally for a random value $x_R \in \{0, 1\}^n$ with $\text{weight}(x_R) = w_x$,

$$\Pr[Z_k = 1, Z_l = 1, x_R] = \sum_{b \in \{0,1\}} \Pr\left[\bigoplus_{i \in S_k \cap S_l} y_i = b, \bigoplus_{i \in S_k \setminus S_l} y_i = 1 - b, \bigoplus_{i \in S_l \setminus S_k} y_i = 1 - b\right]$$

At this point, we diverge from the approach of [4] that computes an upper bound for $\Pr[\bigoplus_{i \in S_k \cap S_l} y_i = b, \bigoplus_{i \in S_k \setminus S_l} y_i = 1 - b, \bigoplus_{i \in S_l \setminus S_k} y_i = 1 - b]$ under the assumption of the *independence of the y_i 's*, which provides consequently an upper bound on $\text{Var}(Z)$.

In fact, we compute precisely $\Pr[Z_k = 1, Z_l = 1]$ through determining the joint PMF of the random variables $(|S_k \cap S_l|, |S_k \setminus S_l|, |S_l \setminus S_k|)$, across two models: (1) the uniform slice model, where x is n -bit vector with fixed weight w_x , and (2) the binomial model where $x \sim \text{Binomial}(n, p_x)$.

The Uniform Slice Model. In this model, the size of the sets S_k and S_l equals w_x . Thus, the size of the sets $S_k \setminus S_l$ and $S_l \setminus S_k$ is entirely determined by the size of $S_k \cap S_l$, namely, $|S_k \setminus S_l| = |S_l \setminus S_k| = w_x - |S_k \cap S_l|$. It suffices then to compute the PMF of the random variable $|S_k \cap S_l|$. The joint probability $\Pr[Z_k = 1, Z_l = 1]$ is given by:

$$\begin{aligned} \Pr[Z_k = 1, Z_l = 1] &= \sum_{\substack{x \in \{0,1\}^n \\ \text{weight}(x) = w_x}} \Pr[Z_k = 1, Z_l = 1, x] \\ &= \sum_{\substack{x \in \{0,1\}^n \\ \text{weight}(x) = w_x}} \sum_{b \in \{0,1\}} \Pr\left[\bigoplus_{i \in S_k \cap S_l} y_i = b, \bigoplus_{i \in S_k \setminus S_l} y_i = 1 - b, \bigoplus_{i \in S_l \setminus S_k} y_i = 1 - b\right] \\ &= \sum_{s=0}^{w_x} \sum_{b \in \{0,1\}} \Pr[|S_k \cap S_l| = s] \Pr\left[\bigoplus_{j=1}^s y_{i_j} = b, \bigoplus_{j=1}^{w_x-s} y_{k_j} = 1 - b, \bigoplus_{j=1}^{w_x-s} y_{l_j} = 1 - b\right] \end{aligned}$$

The Binomial Model. Unlike the uniform slice model, where the size of S_k and S_l is constant, it is now a random variable. Furthermore, we note that $|S_k| = |S_l|$ (both equal to the weight of x). Consequently, we need to compute the joint PMF of the random variables $(|S_k|, |S_k \cap S_l|)$. The joint probability $\Pr[Z_k = 1, Z_l = 1]$ is given by:

$$\begin{aligned} \Pr[Z_k = 1, Z_l = 1] &= \sum_{x \sim \mathcal{B}(n, p_x)} \Pr[Z_k = 1, Z_l = 1, x] \\ &= \sum_{x \sim \mathcal{B}(n, p_x)} \sum_{b \in \{0,1\}} \Pr\left[\bigoplus_{i \in S_k \cap S_l} y_i = b, \bigoplus_{i \in S_k \setminus S_l} y_i = 1 - b, \bigoplus_{i \in S_l \setminus S_k} y_i = 1 - b\right] \\ &= \sum_{r=0}^n \sum_{s=0}^r \sum_{b \in \{0,1\}} \Pr[|S_k| = r, |S_k \cap S_l| = s] \cdot \Pr\left[\bigoplus_{j=1}^s y_{i_j} = b, \bigoplus_{j=1}^{r-s} y_{k_j} = 1 - b, \bigoplus_{j=1}^{r-s} y_{l_j} = 1 - b\right] \end{aligned}$$

The computation of the PMF of $|S_k \cap S_l|$ or the joint PMF of $(|S_k|, |S_k \cap S_l|)$ is sufficiently involved to warrant a separate treatment, which we provide in Section 5 and Appendix 7. For the remainder of this section, we focus on computing, for both the uniform slice and binomial cases, the following probability

$$\Pr\left[\bigoplus_{j=1}^{n_1} y_{i_j} = b_1, \bigoplus_{j=1}^{n_2} y_{k_j} = b_2, \bigoplus_{j=1}^{n_3} y_{l_j} = b_3\right] \quad (3)$$

where

- n_1, n_2, n_3 are nonnegative integers, and $b_1, b_2, b_3 \in \{0, 1\}$,
- the sets $\{y_{i_j} : j \in [1, n_1]\}$, $\{y_{k_j} : j \in [1, n_2]\}$, and $\{y_{l_j} : j \in [1, n_3]\}$ are pairwise disjoint and are subsets of $\{y_i : i \in [0, n-1]\}$.

This probability will directly enable the computation of the variance $\text{Var}(\text{weight}(x \cdot y))$.

4.1 Uniform Slice Model

Theorem 3

Let y be a uniformly random n -bit vector with fixed weight w_y . The probability defined in Equation 3 is given by:

$$\Pr[\bigoplus_{j=1}^{n_1} y_{i_j} = b_1, \bigoplus_{j=1}^{n_2} y_{k_j} = b_2, \bigoplus_{j=1}^{n_3} y_{l_j} = b_3] = \frac{1}{\binom{n}{w_y}} \sum_{\substack{p_1, p_2, p_3 \\ p_1 + p_2 + p_3 \leq w_y}} \prod_{i=1}^3 (p_i \bmod 2 = b_i) \cdot \binom{n_1}{p_1} \binom{n_2}{p_2} \binom{n_3}{p_3} \binom{n - n_1 - n_2 - n_3}{w_y - p_1 - p_2 - p_3} \quad (4)$$

Proof. First, note that the probability space size is $\binom{n}{w_y}$.

Let us now count the number of sequences $(y_0, \dots, y_{n-1})$ that satisfy the conditions of the conjunction $\bigoplus_{j=1}^{n_1} y_{i_j} = b_1, \bigoplus_{j=1}^{n_2} y_{k_j} = b_2, \bigoplus_{j=1}^{n_3} y_{l_j} = b_3$.

Assume for simplicity that $b_1 = b_2 = b_3 = 0$. A sequence that satisfies the aforementioned conjunction is constrained by the following Hamming weight distribution: it must contain an even number p_1 of 1's amongst n_1 positions, an even number p_2 of 1's amongst n_2 positions, an even number p_3 of 1's amongst n_3 positions, and finally $w_y - p_1 - p_2 - p_3$ positions must be set to 1 amongst the remaining $n - n_1 - n_2 - n_3$ positions. The number of such sequences amounts to $\sum_{\substack{\text{even } p_1, p_2, p_3 \\ p_1 + p_2 + p_3 \leq w_y}} \binom{n_1}{p_1} \binom{n_2}{p_2} \binom{n_3}{p_3} \binom{n - n_1 - n_2 - n_3}{w_y - p_1 - p_2 - p_3}$.

For arbitrary b_1, b_2, b_3 , the considered p_1, p_2, p_3 must have the same parity as b_1, b_2, b_3 , respectively. This explains the condition $\prod_{i=1}^3 (p_i \bmod 2 = b_i)$. $\square$

Let $\text{pmf}_U(n, w_x, k, l, s)$ denote the probability $\Pr[|S_k \cap S_l| = s]$, where the probability space is the set of n -bit vectors with weight w_x . Then, computing the variance is straightforward using:

$$\begin{aligned} \text{Var}(Z) &= \sum_{k=0}^{n-1} \text{Cov}(Z_k, Z_k) + 2 \sum_{0 \leq k < l < n} \Pr[Z_k = 1, Z_l = 1] - 2 \sum_{0 \leq k < l < n} \Pr[Z_k = 1] \Pr[Z_l = 1] \\ &= n \cdot \mathbb{E}(Z_k)(1 - \mathbb{E}(Z_k)) + 2 \sum_{0 \leq k < l < n} \Pr[Z_k = 1, Z_l = 1] - n(n-1)\mathbb{E}^2(Z_k) \end{aligned}$$

Algorithm 1: Computation of the variance of $\text{weight}(x \cdot y)$ in the uniform slice model

Input: Nonnegative integers n, w_x, w_y satisfying $w_x, w_y \leq n$.

Random n -bit vectors $(x_0, \dots, x_{n-1})$ and $(y_0, \dots, y_{n-1})$ of weights w_x and w_y respectively.

Output: $\text{Var}(\text{weight}(x \cdot y))$

```

1 tmp ←  $\sum_{s=0}^{\min(w_x, w_y)-1} (-2)^s \frac{\binom{w_x}{s+1} \binom{w_y}{s+1}}{\binom{n}{s+1}};$ 
2 Cov1 ← tmp(1 - tmp);
3 Cov2 ←  $\frac{1}{\binom{n}{w_y}} \sum_{l=1}^{n-1} \sum_{k=0}^{l-1} \sum_{s=0}^{w_x} \sum_{b=0}^1 \text{pmf}_U(n, w_x, k, l, s) \sum_{\substack{p_1, p_2, p_3 \\ p_1 + p_2 + p_3 \leq w_y}} (p_1 \bmod 2 = b_1)(p_2 \bmod 2 = 1 - b)(p_3 \bmod 2 = 1 - b) \binom{s}{p_1} \binom{w_x - s}{p_2} \binom{w_x - s}{p_3} \binom{n - 2w_x + s}{w_y - p_1 - p_2 - p_3}$ 
4 return  $n\text{Cov}_1 + 2\text{Cov}_2 - n(n-1)\text{tmp}^2;$ 

```

4.2 Binomial Model

Lemma 2. Let $y = (y_0, \dots, y_{n-1})$ be an n -bit vector with Bernoulli(p_y) coordinates. Consider further a bit $b \in \{0, 1\}$ and a nonnegative integer $N \leq n$. Then

$$\Pr[\bigoplus_{j=1}^N y_{i_j} = b] = \begin{cases} (1 + (1 - 2p_y)^N)/2 & \text{if } b = 0 \\ (1 - (1 - 2p_y)^N)/2 & \text{otherwise} \end{cases} \quad (5)$$

Proof. Suppose that $b = 0$. Then

$$\begin{aligned} \Pr[\bigoplus_{j=1}^N y_{i_j} = 0] &= \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 0, y_{i_N} = 0] + \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 1, y_{i_N} = 1] \\ &= (1 - p_y) \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 0 \mid y_{i_N} = 0] + p_y \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 1 \mid y_{i_N} = 1] \\ &= (1 - p_y) \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 0] + p_y \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 1] \\ &= (1 - p_y) \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 0] + p_y (1 - \Pr[\bigoplus_{j=1}^{N-1} y_{i_j} = 0]) \end{aligned}$$

If we denote $\Pr[\bigoplus_{j=1}^N y_{i_j} = 0] = F(N)$, then F is defined by the recurrence $F(N) = p_y + (1 - 2p_y)F(N-1)$ and the base case $F(0) = 1$. Solving this recurrence for $F(N)$ gives the closed-form expression $\Pr[\bigoplus_{j=1}^N y_{i_j} = 0] = (1 + (1 - 2p_y)^N)/2$.

For $b = 1$, we apply the same approach to define $\Pr[\bigoplus_{j=1}^N y_{i_j} = 1] = G(N)$ using a recursive function G , given by the same recurrence $G(N) = p_y + (1 - 2p_y)G(N-1)$ as F but with the base case $G(0) = 0$. The solution to this recurrence yields the desired expression. $\square$

Corollary 2. Let $y = (y_0, \dots, y_{n-1})$ be an n -bit vector with Bernoulli(p_y) coordinates. The probability defined in Equation 3 is given by:

$$\Pr[\bigoplus_{j=1}^{n_1} y_{i_j} = b_1, \bigoplus_{j=1}^{n_2} y_{k_j} = b_2, \bigoplus_{j=1}^{n_3} y_{l_j} = b_3] = \prod_{i=1}^3 [1 + (-1)^{b_i} (1 - 2p_y)^{n_i}] / 2 \quad (6)$$

Proof. Follows directly from Lemma 2 and the independence of the y_i 's. $\square$

The algorithm for computing the variance $\text{Var}(\text{weight}(x \cdot y))$ is straightforward, given the probability $\text{pmf}_B(n, p_x, k, l, r, s) = \Pr[|S_k| = r, |S_k \cap S_l| = s]$, defined over the space of n -bit vectors with i.i.d. Bernoulli(p_x) coordinates.

Algorithm 2: Computation of the variance of $\text{weight}(x \cdot y)$ in the binomial model

Input: Positive integer n and real numbers $p_x, p_y \in [0, 1]$.

Random n -bit vectors $(x_0, \dots, x_{n-1}) \sim B(n, p_x)$ and $(y_0, \dots, y_{n-1}) \sim B(n, p_y)$

Output: $\text{Var}(\text{weight}(x \cdot y))$

```

1 tmp ← (1 - (1 - 2p_x p_y)^n) / 2 ;
2 Cov_1 ← tmp(1 - tmp);
3 Cov_2 ←
  1/8 ∑_{l=1}^{n-1} ∑_{k=0}^{l-1} ∑_{r=0}^n ∑_{s=0}^r ∑_{b=0}^1 pmf_B(n, p_x, k, l, r, s) [1 + (-1)^b (1 - 2p_y)^s] [1 + (-1)^{1-b} (1 - 2p_y)^{r-s}]^2 ;
4 return nCov_1 + 2Cov_2 - n(n-1)tmp^2;
```

4.3 Validation and Comparative Analysis

Experimental Validation. We experimentally validate the derived variance expressions through 100,000 trials for two distinct distributions;

- **Uniform slice:** Both polynomials are sampled uniformly from the set of polynomials with degrees less than $n = 250$ and weight w less than $n/2$.
- **Binomial:** Both polynomials have degrees less than $n = 250$ and coefficients drawn independently from a Bernoulli distribution with parameter $\frac{w}{n}$ less than $1/2$.

In each trial, we compute the product modulo $X^n + 1$ and record the weight Z and its square Z^2 . The empirical variance is $\langle Z^2 \rangle - \langle Z \rangle^2$, where $\langle \cdot \rangle$ denotes the sample mean over all trials. As illustrated in Figure 3, the close alignment between the empirical and theoretical variance values validates the accuracy of the analytical formulae.

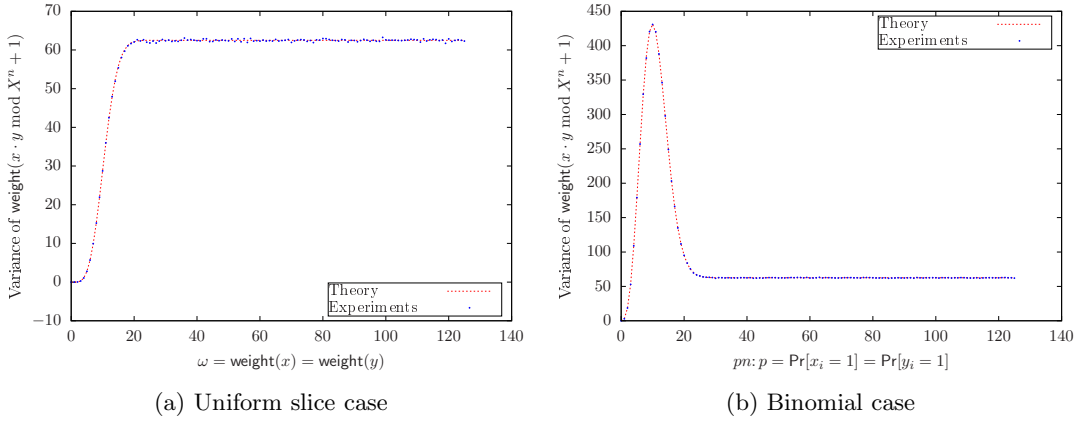


Fig. 3: Variance of the weight of polynomials product modulo $X^n + 1$

Comparison with Existing Approximations. We compare our analytical expressions for the variance with existing approximations from the literature [3,4] in the binomial setting.

Figure 4 plots the variance of the weight of $x \cdot y$, for $n = 250$, and independent vectors x, y with $\text{Bernoulli}(\frac{w}{n})$ coefficients. The curve labeled “**HQC’s approximation**” in Figure 4a plots the approximation from [3], of the variance under the same model but with the simplifying assumption that $\text{weight}(x \cdot y)$ itself follows a binomial distribution. The curve labeled “**Kawachi’s approximation**” in Figure 4b plots the upper bound from [4], on the variance under the same model.

We observe that the approximation from [3] is accurate only for large Bernoulli parameters, specifically for $\frac{w}{n} \in [1/9, 1/2]$. In contrast, the bound from [4] is accurate only for extremely small parameters $\frac{w}{n}$, despite being stated for $\frac{w}{n}$ of order $O(1/\sqrt{n})$.

Tail Probability of $\text{weight}(x \cdot y)$. We compute bounds on the tail probability of $\text{weight}(x \cdot y)$ using Chebyshev’s inequality and compare them with existing bounds from the literature [3,4].

Figure 5 presents the upper bound on the probability that $\text{weight}(x \cdot y) \notin (1 \pm 0.1)\mathbb{E}(\text{weight}(x \cdot y))$ for $n = 250$ and $w \in [0, 125]$, where (1) $w = \text{weight}(x) = \text{weight}(y)$ in the uniform slice model, and (2) $x \sim \text{Binomial}(n, \frac{w}{n})$ and $y \sim \text{Binomial}(n, \frac{w}{n})$ in the binomial model. It also depicts the bounds obtained from existing approximations [3,4] for the binomial setting.

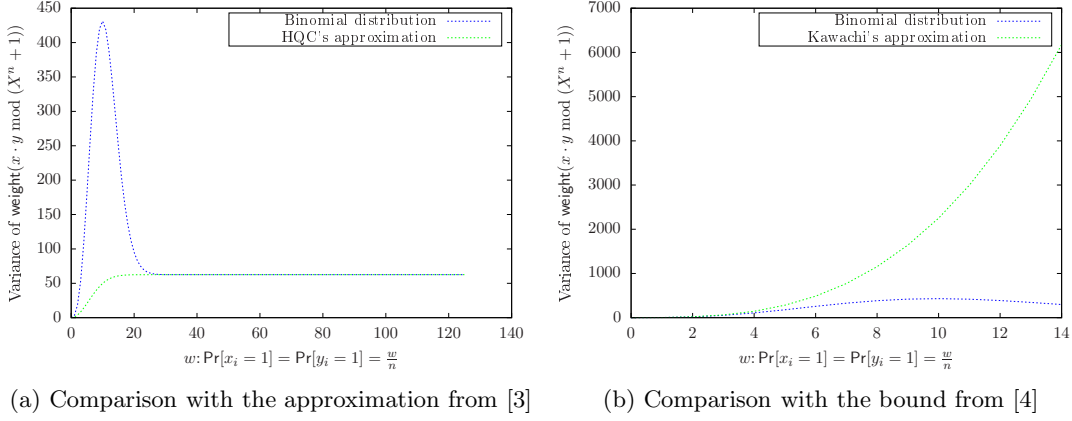


Fig. 4: Comparison of the variance with approximations from the literature [3,4]

We observe that $\text{weight}(x \cdot y)$ is tightly concentrated around its mean in the the uniform slice model. This behavior, however, only holds for large Bernoulli parameters in the binomial setting. Regarding existing approximations, the bound from [3] is reasonable for large Bernoulli parameters, whereas [4] produces entirely irrelevant bounds.

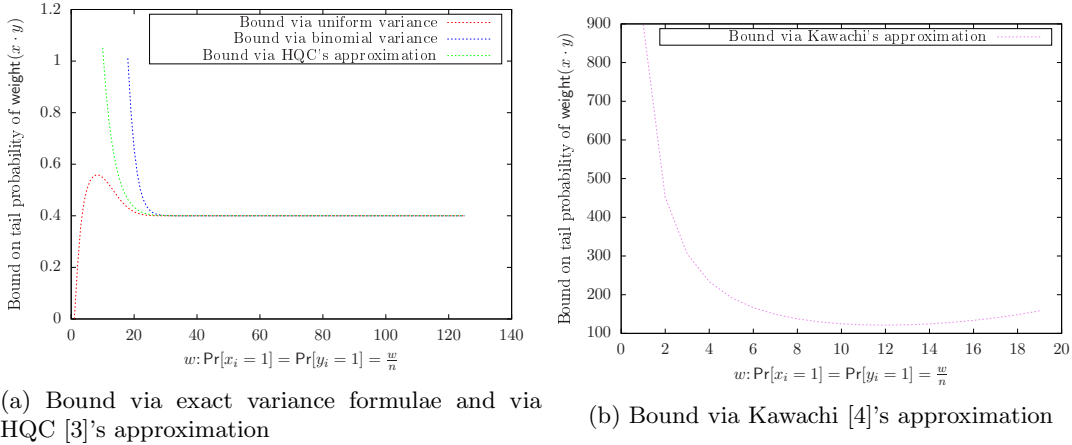


Fig. 5: Upper bound on the tail probability $\Pr[\text{weight}(x \cdot y) \notin (1 \pm 0.1)\mathbb{E}(\text{weight}(x \cdot y))]$ for $n = 250$ and $w \in [0, 125]$

5 Probability Law of $|S_k \cap S_l|$ in the Uniform Slice Model

Let n be a positive integer. Let further w_x, k, l be integers in $[0, n]$ with $k < l < n$. In this section, we will determine the probability mass function of the random variable $|S_k \cap S_l|$, across the probability space of w_x -weighted n -bit vectors $x = (x_0, \dots, x_{n-1})$, where $S_j = \{(j - i) \bmod n : x_i = 1\}$, $j = k, l$.

$$\begin{aligned}
 e \in S_k \cap S_l &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } e = (k - i) \bmod n \text{ and } e = (l - j) \bmod n \\
 &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } \{k - i = l - j \text{ or } k - i + n = l - j \text{ or } k - i = l - j + n\} \\
 &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } \{k - i = l - j \text{ or } k - i + n = l - j\} \\
 &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } \{j - i = l - k \text{ or } i - j = n + k - l\}
 \end{aligned}$$

The penultimate equivalence follows from $i, j \in \mathbb{Z}_n$ and $k < l$, thus $j - i$ cannot equal $n + l - k$.

We conclude that if $|S_k \cap S_l|$ equals a given value $s \in [0, w_x]$, then there exist exactly s pairs (i, j) among the following pairs:

$$\{(0, l - k), \dots, (n + k - l - 1, n - 1)\} \cup \{(n + k - l, 0) \dots (n - 1, l - k - 1)\}$$

satisfying $x_i = x_j = 1$. In fact, a pair (i, j) , for $0 \leq i, j \leq n - 1$, leading to a common element of S_k and S_l must satisfy $x_i = x_j = 1$ in addition to $j - i = l - k$ or $i - j = n + k - l$. We need to distinguish two cases: (1) Case ' $l - k = n + k - l$ ' and (2) case ' $l - k \neq n + k - l$ '.

5.1 The Case ' $l - k = n + k - l$ '

Theorem 4

Given a random n -bit vector $x \in \{0, 1\}^n$ with weight w_x . For nonnegative integers k and l satisfying $k < l < n$ and $l - k = n + k - l$, the probability mass function of the random variable $|S_k \cap S_l|$ is given by

$$\Pr[|S_k \cap S_l| = s] = \begin{cases} 0 & \text{if } s \bmod 2 = 1 \\ \frac{1}{\binom{n}{w_x}} \sum_{w=0}^{w_x} \binom{l-k}{s/2} \binom{l-k-s/2}{w-s/2} \binom{l-k-w}{w_x-w-s/2} & \text{otherwise} \end{cases} \quad (7)$$

Proof. Recall that computing the PMF of $|S_k \cap S_l|$ amounts to computing the probability that there exist exactly s pairs (i, j) amongst the possible pairs

$$\{(0, l - k), \dots, (l - k - 1, n - 1)\} \cup \{(l - k, 0) \dots (n - 1, l - k - 1)\}$$

with $x_i = x_j = 1$, where the probability space is the set of random n -bit vectors $x \in \{0, 1\}^n$ with weight w_x . Thus, the size of the probability space is $\binom{n}{w_x}$.

First note that if (i, j) is possible pair, i.e., a pair leading to an element in $S_k \cap S_l$, then $x_i = x_j = 1$ and $|i - j| = l - k = n + k - l$. It follows that $|j - i| = l - k = n + k - l$, leading to (j, i) being also a possible pair. Therefore, if s is odd, then the probability must be zero.

Suppose now that s is even. We need to specify the weight composition of $x = (x_0, \dots, x_{n-1})$ that gives exactly $s/2$ pairs (i, j) satisfying: (i) $i \in [0, l - k - 1]$, (ii), $j = i + l - k$, and (iii) $x_i = x_j = 1$. In fact, this will give a total of s pairs (I, J) with $I, J \in [0, n - 1]$, $|I - J| = l - k$, and $x_I = x_J = 1$.

Suppose that the weight of the subvector $(x_0, \dots, x_{l-k-1})$ is w . Then we need to set $s/2$ coordinates in $[0, l - k - 1]$ to 1, then, choose $w - s/2$ amongst the remaining $l - k - s/2$ coordinates and set them to 1.

It remains now to describe the weight composition of the second subvector $(x_{l-k}, \dots, x_{n-1})$. Let $i_1, \dots, i_w$ be the indices of the coordinates set to 1 in the subvector $(x_0, \dots, x_{l-k-1})$. Recall that j indices of the pairs (i, j) are constrained by the rule $j = l - k + i$. Thus, the coordinates in $(x_{l-k}, \dots, x_{n-1})$ whose indices are $i_1 + l - k, \dots, i_w + l - k$ will be assigned as follows: $s/2$ coordinates (coresponding to the first chosen $s/2$ indices) are set to 1 and the remaining $w - s/2$ coordinates are set to 0. In this way, we get exactly $s/2$ pairs (i, j) satisfying the above three requirements. It remains now to complete setting the coordinates of the subvector $(x_{l-k}, \dots, x_{n-1})$ to get the partial weight $w_x - w$. To do so, we need to discard w indices from $[l - k, n - 1]$, namely the $i_1 + l - k, \dots, i_w + l - k$ indices, then choose $w_x - w - s/2$ indices and set the corresponding coordinates to 1. $\square$

5.2 The Case ' $l - k \neq n + k - l$ '

In this case, we give a graphic representation to our problem.

Definition 1 (Graph G). Let $(x_0, \dots, x_{n-1}) \in \{0, 1\}^n$ be a random n -bit vector with weight w_x . We define G to be the undirected graph on the vertices $\{x_0, \dots, x_{n-1}\}$ where adjacency is given by the rule: vertex x_i is adjacent to vertex x_j if and only if $(i - j) \bmod n \in \{l - k, n + k - l\}$, which is equivalent to $(i - j) \equiv l - k \pmod{n}$ or $(i - j) \equiv k - l \pmod{n}$.

Note that according to our definition of adjacency, x_i is adjacent to x_j if and only if x_j is adjacent to x_i . We are then consistent with the requirement (or definition) of an undirected graph.

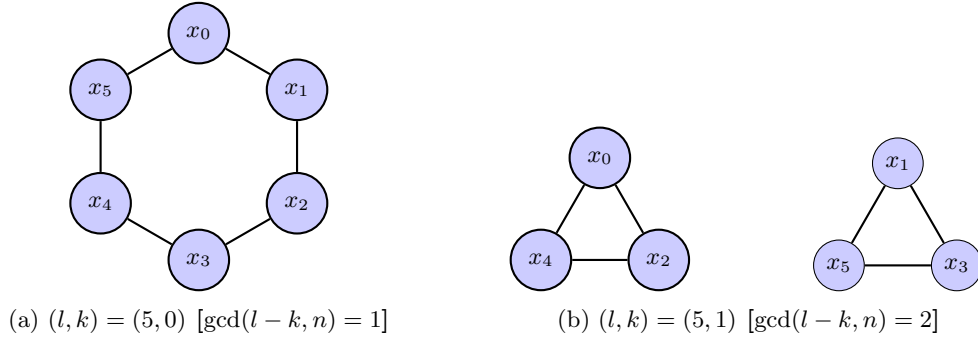


Fig. 6: Graph G for $n = 6$ and different values of (l, k)

We can then reduce the problem of computing the probability that $|S_k \cap S_l|$ takes a given value s , to computing the probability that the graph G contains exactly s edges incident to two vertices both assigned the value 1.

At this stage, we state and prove the following result. The shape of the graph G varies according to the value of $\gcd(l - k, n)$. More precisely, if $\gcd(n, l - k) = 1$, then G is a cycle graph, and if $\gcd(l - k, n) = d$ with $d > 1$, then G is disjoint union of d cycle subgraphs. Figures 6a and 6b exemplify this result for $n = 6$.

Theorem 5

If $\gcd(l - k, n) = 1$, then G is a cycle graph.

Proof. We consider the cycle graph H on the vertices $x_{(i(l-k)) \bmod n}$ for $i = 0 \dots n - 1$, where vertex $x_{(i(l-k)) \bmod n}$ is adjacent to vertex $x_{((i+1)(l-k)) \bmod n}$ for $i = 0 \dots n - 2$, and vertex $x_{((n-1)(l-k)) \bmod n}$ is adjacent to vertex x_0 .

Since $\gcd(l - k, n) = 1$, then $l - k$ is a generator of $(\mathbb{Z}_n, +)$. It follows that the values $(i(l - k)) \bmod n$ span the additive group $\mathbb{Z}_n$. Thus $V(H) = V(G)$.

Moreover, an edge $(x_{(i(l-k)) \bmod n}, x_{((i+1)(l-k)) \bmod n})$, for $i = 0 \dots n - 2$, satisfies $((i + 1)(l - k) - i(l - k)) \bmod n = l - k$. Also, the edge $(x_{((n-1)(l-k)) \bmod n}, x_0)$ satisfies $((n - 1)(l - k)) \bmod n = n + k - l$. Thus $E(H) \subseteq E(G)$.

Let now (x_i, x_j) be an edge in G . Then $(i - j) \bmod n \in \{l - k, n + k - l\}$. Since $l - k$ is a generator of $\mathbb{Z}_n$, then there exists $s \in \mathbb{Z}_n$ such that $i \equiv s(l - k) \pmod{n}$. It follows that $s(l - k) \equiv l - k + j \pmod{n}$ or $s(l - k) \equiv k - l + j \pmod{n}$. Thus $j \equiv (s - 1)(l - k) \pmod{n}$ or $j \equiv (s + 1)(l - k) \pmod{n}$. Which implies $(x_i, x_j) \in E(H)$. We conclude that $E(G) = E(H)$, and thus $G = H$. $\square$

Theorem 6

If $\gcd(l - k, n) = d$, then G is a disjoint union of cycle graphs, each of length n/d .

Proof. Let H_j , for $0 \leq j \leq d-1$, be the cycle graph on the vertices $x_{(j+i(l-k)) \bmod n}$ for $0 \leq i \leq n/d-1$. Concerning the edges of H_j , we build them as follows. Vertex $x_{(j+i(l-k)) \bmod n}$ is adjacent to vertex $x_{(j+(i+1)(l-k)) \bmod n}$ for $0 \leq i \leq n/d-2$, and vertex $x_{(j+(n/d-1)(l-k)) \bmod n}$ is adjacent to vertex $x_{j \bmod n}$.

We will show that G is the disjoint union of the graphs H_j .

First, note that n/d is the order of the subgroup generated by $(l-k) \bmod n$, thus H_j has exactly n/d vertices. Let us prove now that the H_j are pairwise disjoint. Suppose there is a common vertex in H_j and $H_{j'}$, then there exist $i, i' \in [0, n/d-1]$ such that $(j+i(l-k)) \bmod n = (j'+i'(l-k)) \bmod n$. Thus $j-j' - (i'-i)(l-k) \equiv 0 \pmod{n}$. This implies that $j-j'$ is a multiple of d (both n and $l-k$ are multiples of d) which implies $j=j'$ as $0 \leq j, j' \leq d-1$. Thus the H_j are pairwise disjoint, and their union $\bigcup_j H_j$ has cardinality $d \cdot n/d = n$. Since $V(\bigcup_j H_j) \subseteq V(G)$ and G has n vertices, we conclude that $V(G) = V(\bigcup_j H_j)$.

Moreover, let $(x_r, x_{r'})$ be an edge in G . Then $r'-r \equiv l-k \pmod{n}$ or $r'-r \equiv k-l \pmod{n}$. Moreover, there exist j, i , where $0 \leq j \leq d-1$ and $0 \leq i \leq n/d-1$, such that $x_r \in H_j$ and $r \equiv j+i(l-k) \pmod{n}$. Thus $r' \equiv j+i(l-k) + (l-k) \pmod{n}$ or $r' \equiv j+i(l-k) + (k-l) \pmod{n}$. It follows that $r' \equiv j+(i+1)(l-k) \pmod{n}$ or $r' \equiv j+(i-1)(l-k) \pmod{n}$. Which means $(x_r, x_{r'})$ is an edge of H_j , and $E(G) \subseteq E(\bigcup_j H_j)$.

Let now $(x_r, x_{r'}) \in E(\bigcup_j H_j)$. Since the H_j 's are pairwise disjoint, then there exists $j \in [0, d-1]$ such that $(x_r, x_{r'}) \in H_j$. It follows that there exists $i \in [0, n/d-1]$ such that $r \equiv j+i(l-k) \pmod{n}$ and $r' \equiv j+(i+1)(l-k) \pmod{n}$. This implies $r'-r \equiv l-k \pmod{n}$. Which means $(x_r, x_{r'}) \in E(G)$. Therefore, $E(\bigcup_j H_j) = E(G)$.

We conclude that $G = \bigcup_j H_j$. □

Recall that the goal of this subsection is to determine the probability that the graph G contains exactly a given number, say s , of edges whose endpoints are both set to 1. Since we have demonstrated that G is either a cycle graph, or a disjoint union of cycle graphs, we first compute this probability for a cycle graph, then expand the analysis to treat the more general case of a disjoint union of cycle graphs.

5.2.1 Cycle Graphs

Throughout this subsection, i, j, s will denote nonnegative integers, satisfying $0 \leq i \leq n$ and $0 \leq j, s \leq w_x$.

Definition 2 (Cycle-PMF). For $i > 0$, consider a random i -bit vector $(x_0, \dots, x_{i-1}) \in \{0, n\}^i$ with fixed weight j .

$\text{cycle-pmf}(i, j, s)$ denotes the probability that the cycle graph $x_0 - \dots - x_{i-1} - x_0$ contains exactly s edges with both endpoints equal to 1. The probability is taken over all random i -bit vectors $(x_0, \dots, x_{i-1})$ with weight j .

To compute $\text{cycle-pmf}(i, j, s)$, we assign a value to one vertex in the cycle graph $x_0 - \dots - x_{i-1} - x_0$, say x_0 . It suffices then to compute the probability that the linear chain $x_1 - \dots - x_{i-1}$ contains exactly s or $s-1$ edges incident to vertices both equal to 1. We introduce now some auxiliary functions that compute the aforementioned probabilities.

Definition 3 (Functions f_0, f_1, g, f). Let i, j be nonnegative integers. Let further $x_0 - \dots - x_i$ be a linear chain, where $(x_0, \dots, x_i)$ is an $(i+1)$ -bit random vector of weight j .

$f_0(i, j, s)$ denotes the conditional probability that the linear chain $x_0 - \dots - x_i$ contains exactly s edges with both endpoints equal to 1, given that x_0 is set to 0.

$f_1(i, j, s)$ denotes the conditional probability that $x_0 - \dots - x_i$ contains exactly s edges with both endpoints equal to 1, given that x_0 is set to 1.

$g(i, j, s, b, c)$, for $b, c \in \{0, 1\}$, denotes the conditional probability that $x_0 - \dots - x_i$ contains exactly s edges with both endpoints equal to 1, given that x_0 and x_i are set to b and c respectively.

$f(i, j, s)$ denotes the probability that $x_0 - \dots - x_i$ contains exactly s edges with both endpoints equal to 1.

Let us now proceed to the computation of the above functions. Concerning the functions f_0 , f_1 , and g , we employ a dynamic programming approach for their computation. This involves first evaluating the functions at their base cases (initial argument values), then providing a recurrence relation that allows to iteratively evaluate the functions at all further values.

Form. 1. and **Form. 2.** give respectively the formulae for computing f_0 , f_1 , and g ; the base cases are immediate, whilst the recurrence relations follow from:

$$\begin{aligned} f_0(i, j, s) &= \Pr[x_1 = 0 \mid x_0 = 0]f_0(i-1, j, s) + \Pr[x_1 = 1 \mid x_0 = 0]f_1(i-1, j, s) \\ f_1(i, j, s) &= \Pr[x_1 = 0 \mid x_0 = 1]f_0(i-1, j-1, s) + \Pr[x_1 = 1 \mid x_0 = 1]f_1(i-1, j-1, s-1) \\ g(i, j, s, b, c) &= \Pr[x_1 = 0 \mid x_0 = b, x_i = c]g(i-1, j-b, s, 0, c) + \Pr[x_1 = 1 \mid x_0 = b, x_i = c]g(i-1, j-b, s-b, 1, c) \end{aligned}$$

Where

$$\begin{aligned} \Pr[x_1 = 0 \mid x_0 = 0] &= \frac{\binom{i-1}{j}}{\binom{i}{j}} & \text{and } \Pr[x_1 = 1 \mid x_0 = 0] &= \frac{\binom{i-1}{j-1}}{\binom{i}{j}} \\ \Pr[x_1 = 0 \mid x_0 = 1] &= \frac{\binom{i-1}{j-1}}{\binom{i}{j-1}} & \text{and } \Pr[x_1 = 1 \mid x_0 = 1] &= \frac{\binom{i-1}{j-2}}{\binom{i}{j-1}} \\ \Pr[x_1 = 0 \mid x_0 = b, x_i = c] &= \frac{\binom{i-2}{j-b-c}}{\binom{i-1}{j-b-c}} & \text{and } \Pr[x_1 = 1 \mid x_0 = b, x_i = c] &= \frac{\binom{i-2}{j-b-c-1}}{\binom{i-1}{j-b-c}} \end{aligned}$$

$f_0(i, j, s) = f_1(i, j, s) = 0$ if $(j > i + 1) \vee (s > i) \vee (s > j - 1) \vee (s < 0)$. Otherwise:			
$f_0(i, j, s) =$	$(j = 0) \cdot (s = 0)$		if $i = 0$
	$(j \leq i) \cdot (s = 0)$		if $i = 1$
	$\begin{cases} 0 & \text{if } \binom{i}{j} = 0 \\ \frac{\binom{i-1}{j}}{\binom{i}{j}} f_0(i-1, j, s) + \frac{\binom{i-1}{j-1}}{\binom{i}{j}} f_1(i-1, j, s) & \text{otherwise} \end{cases}$		otherwise
	$\begin{cases} (j = 1) \cdot (s = 0) & \text{if } i = 0 \\ (j = 1) \cdot (s = 0) + (j = 2) \cdot (s = 1) & \text{if } i = 1 \\ 0 & \text{if } \binom{i}{j-1} = 0 \\ \frac{\binom{i-1}{j-1}}{\binom{i}{j-1}} f_0(i-1, j-1, s) + \frac{\binom{i-1}{j-2}}{\binom{i}{j-1}} f_1(i-1, j-1, s-1) & \text{otherwise} \end{cases}$		otherwise

Form. 1. Computation of f_0 and f_1

$g(i, j, s, b, c) = 0$ if $(j > i + 1) \vee (s > i) \vee (j > b + c) \vee (s < 0) \vee (i = 0)$. Otherwise:			
$g(i, j, s, b, c) =$	$(j = b + c) \cdot (s = b \cdot c)$		if $i = 1$
	$\begin{cases} 0 & \text{if } \binom{i-1}{j-b-c} = 0 \\ \frac{\binom{i-2}{j-b-c}}{\binom{i-1}{j-b-c}} g(i-1, j-b, s, 0, c) + \frac{\binom{i-2}{j-b-c-1}}{\binom{i-1}{j-b-c}} g(i-1, j-b, s-b, 1, c) & \text{otherwise} \end{cases}$		otherwise

Form. 2. Computation of g

Form. 3. is justified by

$$f(i, j, s) = \Pr[x_0 = 0]f_0(i, j, s) + \Pr[x_0 = 1]f_1(i, j, s)$$

where $\Pr[x_0 = 0]$ and $\Pr[x_0 = 1]$ are given by $\frac{\binom{i}{j}}{\binom{i+1}{j}}$ and $\frac{\binom{i}{j-1}}{\binom{i+1}{j}}$ respectively.

$$f(i, j, s) = \begin{cases} 0 & \text{if } \binom{i+1}{j} = 0 \\ \frac{\binom{i}{j}}{\binom{i+1}{j}}f_0(i, j, s) + \frac{\binom{i}{j-1}}{\binom{i+1}{j}}f_1(i, j, s) & \text{otherwise} \end{cases}$$

Form. 3. Computation of f

Finally, for cycle-pmf given in **Form. 4.** , it is clear that if $i \leq 2$, then $\text{cycle-pmf}(i, j, s) = f(i-1, j, s)$, otherwise

$$\begin{aligned} \text{cycle-pmf}(i, j, s) &= \Pr[x_0 = 0]f(i-2, j, s) + \sum_{b, c \in \{0,1\}} \Pr[x_0 = 1, x_1 = b, x_{i-1} = c] \\ &\quad g(i-2, j-1, s-b-c, b, c) \end{aligned}$$

where $\Pr[x_0 = 0]$ in case of the cycle graph considered in cycle-pmf^1 is given by $\frac{\binom{i-1}{j}}{\binom{i}{j}}$, whilst $\Pr[x_0 = 1, x_1 = b, x_{i-1} = c]$ is given by $\frac{\binom{i-3}{j-1-b-c}}{\binom{i}{j}}$.

$$\text{cycle-pmf}(i, j, s) = \begin{cases} 0 & \text{if } \binom{i}{j} = 0 \\ f(i-1, j, s) & \text{if } i < 3 \\ \frac{\binom{i-1}{j}}{\binom{i}{j}}f(i-2, j, s) + \sum_{b, c \in \{0,1\}} \frac{\binom{i-3}{j-1-b-c}}{\binom{i}{j}}g(i-2, j-1, s-b-c, b, c) & \text{otherwise} \end{cases}$$

Form. 4. Computation of cycle-pmf

Implementation Consideration We precompute all the binomial coefficient $\binom{i}{j}$ for $0 \leq i \leq n$ and $0 \leq j \leq w_x$. Furthermore, we employ a dynamic programming approach to precompute $f_0(i, j, s)$, $f_1(i, j, s)$, and $g(i, j, s, b, c)$ for all nonnegative integers i, j, s , with $0 \leq j, s \leq w_x$ and $0 \leq i \leq n$, and all bits b, c . Following the formulae in **Form. 1.** and **Form. 2.** , we fill the tables dp_{f_0} , dp_{f_1} , and dp_g respectively in a bottom-up manner. This precomputation step, which requires $O(n \cdot w_x^2)$ time and space, allows for a constant time computation of $\text{cycle-pmf}(i, j, s)$, as the result is a simple combination of the stored values.

5.2.2 Disjoint Union of Cycle Graphs

Suppose now that graph G is disjoint union of d cycle graphs $G = \bigcup_{i=0 \dots d-1} G_i$, where $d > 1$ (the case $d = 1$ has been treated in the previous subsection). The vertices of G are again the coordinates of an n -bit random vector $(x_0, \dots, x_{n-1})$ with weight w_x .

¹ The underlying probability space should be inferred from the context. Specifically, the quantity $\Pr[x_0 = 0]$ in case of the cycle graph considered for cycle-pmf is given by $\frac{\binom{i-1}{j}}{\binom{i}{j}}$, whereas the same quantity, when considered for the linear chain underlying f_0 , f_1 , f , or g , is $\frac{\binom{i}{j}}{\binom{i+1}{j}}$.

Let S denote the total number of edges in G that are incident to two vertices both equal to 1. Our goal is to determine the PMF of S .

Let S_i denote the number of edges in G_i , for $0 \leq i \leq d-1$, that are incident to two vertices both equal to 1. Let further $\text{weight}(G_i)$ denote the number of vertices in G_i that are equal to 1. Then

$$\Pr[S = s] = \sum_{\substack{w_0, \dots, w_{d-1} \\ \sum w_i = w_x}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \Pr\left[\bigwedge_{i=0}^{d-1} \text{weight}(G_i) = w_i\right] \cdot \Pr\left[\bigwedge_{i=0}^{d-1} S_i = s_i \mid \bigwedge_{i=0}^{d-1} \text{weight}(G_i) = w_i\right]$$

We know from Theorem 6 that the graphs G_i contain n/d vertices each. It follows that $\Pr[\bigwedge_{i=0}^{d-1} \text{weight}(G_i) = w_i] = \frac{1}{\binom{n}{w_x}} \prod_{i=0}^{d-1} \binom{n/d}{w_i}$. On the other hand, since the graphs G_i are pairwise disjoint, the conditional distributions $S_0 | \text{weight}(G_0), \dots, S_{d-1} | \text{weight}(G_{d-1})$ are mutually independent:

$$\begin{aligned} \Pr\left[\bigwedge_{i=0}^{d-1} S_i = s_i \mid \bigwedge_{i=0}^{d-1} \text{weight}(G_i) = w_i\right] &= \prod_{i=0}^{d-1} \Pr[S_i = s_i | \text{weight}(G_i) = w_i] \\ &= \prod_{i=0}^{d-1} \text{cycle-pmf}(n/d, w_i, s_i) \end{aligned}$$

We obtain the following expression:

$$\Pr[S = s] = \frac{1}{\binom{n}{w_x}} \sum_{\substack{w_0, \dots, w_{d-1} \\ \sum w_i = w_x}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \prod_{i=0}^{d-1} \binom{n/d}{w_i} \text{cycle-pmf}(n/d, w_i, s_i)$$

Form. 5. Computation of the PMF of S

Implementation Consideration. The probability in **Form. 5.** can be computed in polynomial time in n if we employ a dynamic programming approach, as we did previously for the functions f_0 , f_1 , and g .

Let D be the largest divisor of n . Consider the following function h :

$$\begin{aligned} h: [0, D] \times [0, w_x] \times [0, w_x] \times [0, D] &\longrightarrow \mathbb{R} \\ (N, w, s, d) &\longmapsto \sum_{\substack{w_0, \dots, w_{d-1} \\ \sum w_i = w}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \prod_{i=0}^{d-1} \binom{N}{w_i} \text{cycle-pmf}(N, w_i, s_i) \end{aligned}$$

Then

$$\begin{aligned} h(N, w, s, d) &= \sum_{w_0=0}^w \sum_{s_0=0}^s \binom{N}{w_0} \text{cycle-pmf}(N, w_0, s_0) \sum_{\substack{w_1, \dots, w_{d-1} \\ \sum w_i = w - w_0}} \sum_{\substack{s_1, \dots, s_{d-1} \\ \sum s_i = s - s_0}} \prod_{i=1}^{d-1} \binom{N}{w_i} \text{cycle-pmf}(N, w_i, s_i) \\ &= \sum_{w_0=0}^w \sum_{s_0=0}^s \binom{N}{w_0} \text{cycle-pmf}(N, w_0, s_0) h(N, w - w_0, s - s_0, d - 1) \end{aligned} \tag{8}$$

We can then precompute all the values of $h(N, w, s, d)$, for the considered set of parameters, using a dynamic programming table that we fill in bottom-up manner according the recurrence relation defined in Equation 8. This allows the computation of the PMF of S in time and space polynomial in n .

5.3 Piecing all Together

5.3.1 Probability Law of $|S_k \cap S_l|$

We are now ready to state the algorithm that computes the PMF of $|S_k \cap S_l|$, for $0 \leq k < l < n$ and $S_j = \{(j - i) \bmod n : x_i = 1\}$, for $j = k, l$; the probability space is the set of n -bit vectors $x = (x_0, \dots, x_{n-1}) \in \{0, 1\}^n$ of weight w_x .

Algorithm 3: Computation of the PMF of $|S_k \cap S_l|$

Input: Nonnegative integers n, w_x, k, l , and s with $k < l < n$ and $s \leq w_x \leq n$

Output: $\Pr[|S_k \cap S_l| = s] = \text{pmf}_U(n, w_x, k, l, s)$

```

1 if  $n + k - l = l - k$  then
    |  $res = \begin{cases} 0 & \text{if } s \bmod 2 = 1 \\ \frac{1}{\binom{n}{w_x}} \sum_{w=0}^{w_x} \binom{l-k}{s/2} \binom{l-k-s/2}{w-s/2} \binom{l-k-w}{w_x-w-s/2} & \text{otherwise} \end{cases}$ 
2 else
    | // We assume the values  $f_0(i, j, t)$ ,  $f_1(i, j, t)$ ,  $g(i, j, t, b, c)$ , and  $h(N, w, t, p)$ ,
    | // for all parameter sets under consideration, are precomputed
    | // using dynamic programming tables as mentioned earlier
3    $d = \gcd(l - k, n)$ ;
4   if  $d = 1$  then
5   |    $res = \text{cycle-pmf}(n, w_x, s)$ ;      /* cycle-pmf is computed according to Form. 4. */
6   else
7   |    $res = \frac{1}{\binom{n}{w_x}} h(n/d, w_x, s, d)$  ;
8 return res;
```

Theorem 7

Algorithm 3 computes the PMF of $|S_k \cap S_l|$ in time and space polynomial in n . □

5.3.2 Optimized Computation of the Variance

Note that the PMF of $|S_k \cap S_l|$, besides depending on n and w_x , is a function solely of $\gcd(l - k, n)$. Therefore, $\Pr[|S_k \cap S_l| = s] = \Pr[|S_{(k+t) \bmod n} \cap S_{(l+t) \bmod n}| = s]$ for all integers $t \in [0, n-1]$ and all $s \in [0, w_x]$. This property enables a significant simplification in the computation of the variance $\text{Var}(\text{weight}(x \cdot y))$. Preserving the previous notations, let $Z = \text{weight}(x \cdot y) = \sum_{k=0}^n Z_k$, with $Z_k = \bigoplus_{0 \leq i < n} x_i y_{(k-i) \bmod n}$. Then, instead of a nested loop over indices k and l when computing the covariance coefficients, the calculation is reduced to a single loop, as detailed below:

$$\begin{aligned}
\sum_{0 \leq k < l < n} \Pr[Z_k = 1 \wedge Z_l = 1] &= \sum_{0 \leq k < l < n} \sum_{s=0}^{w_w} \sum_{b \in \{0,1\}} \text{pmf}_U(n, w_x, k, l, s) \Pr\left[\bigoplus_{j=1}^s y_{i_j} = b \wedge \bigoplus_{j=1}^{w_x-s} y_{k_j} = 1 - b \wedge \bigoplus_{j=1}^{w_x-s} y_{l_j} = 1 - b\right] \\
&= \frac{n}{2} \sum_{l=1}^n \sum_{s=0}^{w_w} \sum_{b \in \{0,1\}} \text{pmf}_U(n, w_x, 0, l, s) \Pr\left[\bigoplus_{j=1}^s y_{i_j} = b \wedge \bigoplus_{j=1}^{w_x-s} y_{k_j} = 1 - b \wedge \bigoplus_{j=1}^{w_x-s} y_{l_j} = 1 - b\right]
\end{aligned}$$

5.4 Experimental Results

We experimentally validate our formula for $\Pr[|S_k \cap S_l| = s]$ over a sample of 100,000 trials. In each trial, we generate a uniformly random polynomial x with degree less than $n = 250$ and fixed weight $w_x = 65$, then record the size $|S_k \cap S_l|$. The empirical value of $\Pr[|S_k \cap S_l| = s]$ is estimated as the proportion of trials in which $|S_k \cap S_l| = s$.

Since $\Pr[|S_k \cap S_l| = s]$, besides depending on n and w_x , is a function solely of $l - k$ and s , we plot in Figure 7 this PMF for the three values: (i) $l - k = 17$, corresponding to the case ' $\gcd(l - k, n) = 1$ ', (ii) $l - k = 25$, corresponding to the case ' $\gcd(l - k, n) \neq 1$ ', and (iii) $l - k = 125$ corresponding to the case ' $l - k = n + k - l$ '. The strong concordance between the 'Theory' and 'Experiments' curves verifies our theoretical derivations.

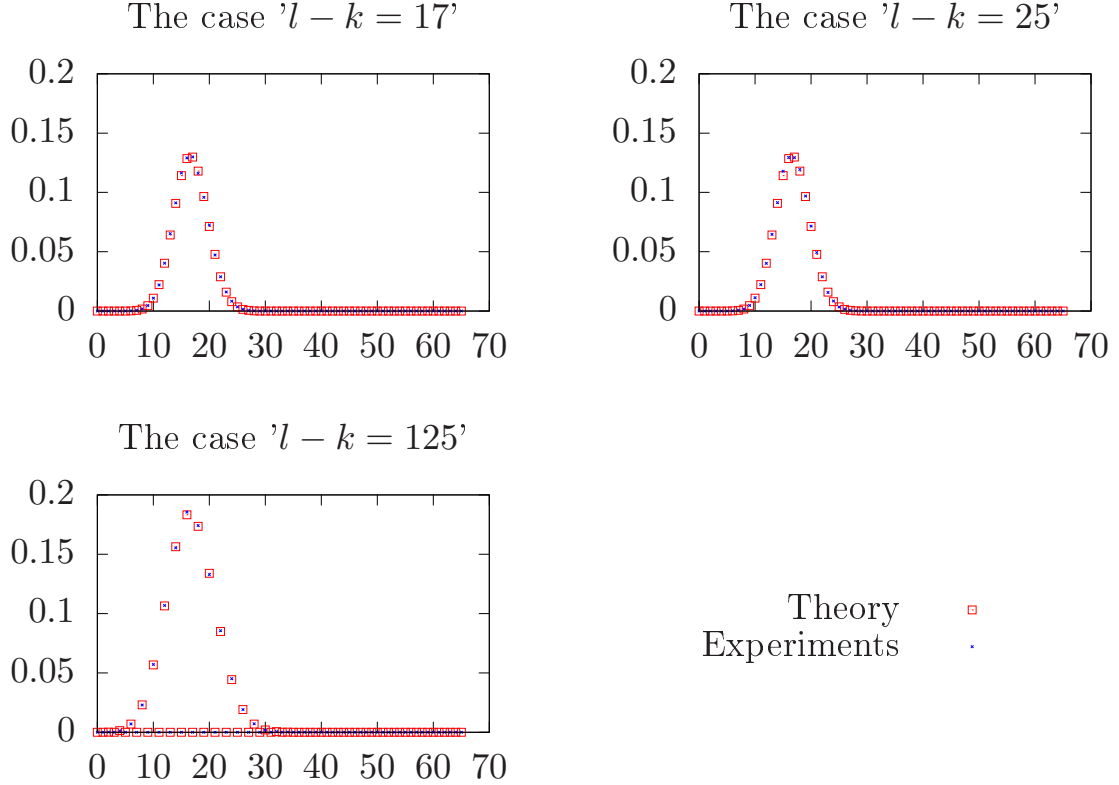


Fig. 7: Probability $\Pr[|S_k \cap S_l| = s]$ for $n = 250$ and $w_x = 65$. The x -axis corresponds to the values $s \in [0, w_x]$

6 Hamming Quasi-Cyclic Codes

We compute in this section the expectation and variance of the weight of the error vector in the HQC scheme [3]. We derive our results in both the uniform slice and binomial models.

We adopt the following convention in this section: $\{v_k\}_{k=0}^{n-1}$ refers to the set of coordinates of vector $v = (v_0, \dots, v_{n-1})$; specifically, $\{x_k\}_{k=0}^{n-1}$ denotes the set of coordinates of x , and $\{r_{2,k}\}_{k=0}^{n-1}$ refers to the coordinates of r_2 , etc. Also, n -bit vectors are identified with polynomials in the ring $\mathbb{F}_2[X]/(X^n + 1)$, and thus their arithmetic (addition and multiplication) is performed modulo $X^n + 1$. Finally, $\mathbb{R}_w$ denotes the set of n -bit vectors with weight w .

6.1 Uniform Slice Model

Definition 4 (HQC's Error Vector–Uniform Model–). Let n be a nonnegative integer, and w, w_1, w_2, w_e be nonnegative integers less or equal than n . The error vector is given by $e' = x \cdot r_2 + y \cdot r_1 + e$, where

- $x, y \in \mathbb{R}_w$.
- $r_1 \in \mathbb{R}_{w_1}$ and $r_2 \in \mathbb{R}_{w_2}$
- $e \in \mathbb{R}_{w_e}$

For convenience, we denote by X and Y the products $x \cdot r_2$ and $r_1 \cdot y$ respectively.

6.1.1 Expectation

The coordinates $\{e'_k\}_{k=0}^{n-1}$ of e' satisfy $e'_k = X_k \oplus Y_k \oplus e_k$. Let $W = \text{weight}(e')$ denote the weight of e' . Then $W = \sum_{k=0}^{n-1} e'_k$. And, by linearity of the expectation, the expectation of $\text{weight}(e')$ is given by

$$\begin{aligned} \mathbb{E}(W) &= \sum_{k=0}^{n-1} \mathbb{E}(e'_k) = \sum_{k=0}^{n-1} \Pr[e'_k = 1] \\ &= \sum_{k=0}^{n-1} \sum_{b,c \in \{0,1\}} \Pr[X_k = b] \Pr[e_k = c] \Pr[Y_k = 1 \oplus b \oplus c] \end{aligned}$$

We have

$$\Pr[e_k = c] = \frac{\binom{n-1}{w_e-c}}{\binom{n}{w_e}} = c \frac{w_e}{n} + (1-c) \frac{n-w_e}{n}$$

Let now E_X denote the expectation of $\text{weight}(X)$, and E_Y be the expectation of $\text{weight}(Y)$. E_X and E_Y are computed according to Corollary 1. We have already established that $E_X = n \Pr[X_k = 1]$ and $E_Y = n \Pr[Y_k = 1]$ for $k = 0, \dots, n-1$. Therefore:

$$\begin{aligned} \Pr[X_k = b] &= b \frac{E_X}{n} + (1-b) \frac{n-E_X}{n} \\ \Pr[Y_k = 1 \oplus b \oplus c] &= (1 \oplus b \oplus c) \frac{E_Y}{n} + (b \oplus c) \frac{n-E_Y}{n} \end{aligned}$$

We notice that $\Pr[e'_k = 1]$ does not depend on k . Thus, $\mathbb{E}(W) = n \Pr[e'_k = 1]$. We derive then the following theorem.

Theorem 8 (Expectation)

The expectation of the error vector in the HQC scheme, as defined in Definition 4, is given by

$$\mathbb{E}(\text{weight}(e')) = \frac{E_Y}{n}(n-w_e)\left(1-\frac{E_X}{n}\right) + \frac{E_X}{n}(n-w_e)\left(1-\frac{E_Y}{n}\right) + w_e\left(1-\frac{E_X}{n}\right)\left(1-\frac{E_Y}{n}\right) + w_e \frac{E_X E_Y}{n^2}$$

where E_X and E_Y denote the expectation of $\text{weight}(X)$ and $\text{weight}(Y)$ respectively, computed via Corollary 1. $\square$

6.1.2 Variance

Let E denote the expectation of $\text{weight}(e')$. We compute the variance of $\text{weight}(e')$ by determining the covariance coefficients $\text{Cov}(e'_k, e'_l)$:

$$\begin{aligned}
\text{Var}(\text{weight}(e')) &= \sum_{0 \leq k, l < n} \text{Cov}(e'_k, e'_l) \\
&= \sum_{k=0}^{n-1} \text{Cov}(e'_k, e'_k) + 2 \sum_{0 \leq k < l < n} \text{Cov}(e'_k, e'_l) \\
&= n \Pr[e'_k = 1](1 - \Pr[e'_k = 1]) - n(n-1)(\Pr[e'_k = 1])^2 + 2 \sum_{0 \leq k < l < n} \Pr[e'_k = 1, e'_l = 1] \\
&= E - E^2 + 2 \sum_{0 \leq k < l < n} \Pr[e'_k = 1, e'_l = 1]
\end{aligned}$$

Let us determine now the joint probability $\Pr[e'_k = 1, e'_l = 1]$.

$$\begin{aligned}
\Pr[e'_k = 1, e'_l = 1] &= \Pr[X_k \oplus Y_k \oplus e_k = 1, X_l \oplus Y_l \oplus e_l = 1] \\
&= \sum_{b, b', c, c' \in \{0,1\}} \Pr[X_k = b, Y_k = c, e_k = 1 \oplus b \oplus c, X_l = b', Y_l = c', e_l = 1 \oplus b' \oplus c'] \\
&= \sum_{b, b', c, c' \in \{0,1\}} \Pr[(X_k = b, X_l = b'), (Y_k = c, Y_l = c'), (e_k = 1 \oplus b \oplus c, e_l = 1 \oplus b' \oplus c')] \\
&= \sum_{b, b', c, c' \in \{0,1\}} \Pr[X_k = b, X_l = b'] \Pr[Y_k = c, Y_l = c'] \Pr[e_k = 1 \oplus b \oplus c, e_l = 1 \oplus b' \oplus c']
\end{aligned}$$

The probability $\Pr[e_k = 1 \oplus b \oplus c, e_l = 1 \oplus b' \oplus c']$ is straightforward and is equal to $\frac{\binom{n-2}{w_e - (1 \oplus b \oplus c) - (1 \oplus b' \oplus c')}}{\binom{n}{w_e}}$. For $\Pr[X_k = b, X_l = b']$ and $\Pr[Y_k = c, Y_l = c']$, we proceed as follows.

$$\begin{aligned}
\Pr[X_k = b, X_l = b'] &= \Pr\left[\bigoplus_{i+j \equiv k \pmod{n}} x_i r_{2,j} = b, \bigoplus_{i+j \equiv l \pmod{n}} x_i r_{2,j} = b'\right] \\
&= \sum_{r_2 \in \mathbb{R}_{w_2}} \Pr\left[\bigoplus_{i \in S_k} x_i = b, \bigoplus_{i \in S_l} x_i = b'\right] \\
&= \sum_{s=0}^{w_2} \sum_{d \in \{0,1\}} \text{pmf}_U(n, w_2, k, l, s) \Pr\left[\bigoplus_{j=1}^s x_{i_j} = d, \bigoplus_{j=1}^{w_2-s} x_{k_j} = b \oplus d, \bigoplus_{j=1}^{w_2-s} x_{l_j} = b' \oplus d\right]
\end{aligned} \tag{9}$$

The probability $\text{pmf}_U(n, w_2, k, l, s)$ is computed according to Algorithm 3 over the probability space of n -bit vectors with weight w_2 . Whereas, the probability $\Pr[\bigoplus_{j=1}^s x_{i_j} = d, \bigoplus_{j=1}^{w_2-s} x_{k_j} = b \oplus d, \bigoplus_{j=1}^{w_2-s} x_{l_j} = b' \oplus d]$ can be computed using Theorem 3, over the sample space of n -bit vectors with weight w .

Using the same argument, we get

$$\Pr[Y_k = c, Y_l = c'] = \sum_{s=0}^{w_1} \sum_{d \in \{0,1\}} \text{pmf}_U(n, w_1, k, l, s) \Pr\left[\bigoplus_{j=1}^s y_{i_j} = d, \bigoplus_{j=1}^{w_1-s} y_{k_j} = c \oplus d, \bigoplus_{j=1}^{w_1-s} y_{l_j} = c' \oplus d\right] \tag{10}$$

Therefore, the variance of the weight of the error vector e' is given by the following theorem.

Theorem 9 (Variance)

Consider the error vector e' in the HQC scheme, defined in Definition 4. Let further E denote the expectation of $\text{weight}(e')$. The variance of $\text{weight}(e')$ is given by:

$$\text{Var}(\text{weight}(e')) = E - E^2 + 2 \sum_{0 \leq k < l < n} \sum_{b, b', c, c' \in \{0,1\}} \frac{\binom{n-2}{w_e - (1 \oplus b \oplus c) - (1 \oplus b' \oplus c')}}{\binom{n}{w_e}} \Pr[X_k = b, X_l = b'] \Pr[Y_k = c, Y_l = c']$$

where $\Pr[X_k = b, X_l = b']$ and $\Pr[Y_k = c, Y_l = c']$ are given by Equations 9 and 10 respectively. $\square$

6.2 Binomial Case

The error vector in the HQC scheme, for the binomial case, is defined as follows.

Definition 5 (HQC's Error Vector–Binomial case–). Let n be a nonnegative integer, and p, p_1, p_2, p_e be real numbers in $[0, 1]$. The error vector is given by $e' = x \cdot r_2 + y \cdot r_1 + e$, where

- $x, y \sim B(n, p)$
- $r_1 \sim B(n, p_1)$ and $r_2 \sim B(n, p_2)$
- $e \sim B(n, p_e)$

We maintain the same notation, as in Definition 4, namely $X = x \cdot r_2$ and $Y = r_1 \cdot y$.

We proceed now to the computation of the expectation and variance of $\text{weight}(e')$.

Theorem 10 (Expectation)

Let e' be the error vector, as described in Definition 5. The expectation $\mathbb{E}(\text{weight}(e'))$ is given by

$$\mathbb{E}(\text{weight}(e')) = n \sum_{b, c \in \{0,1\}} \Pr[X_k = b] \Pr[e_k = c] \Pr[Y_k = 1 \oplus b \oplus c]$$

Where

$$\begin{aligned} \Pr[X_k = b] &= \frac{b}{2}(1 - (1 - 2pp_2)^n) + \frac{1-b}{2}(1 + (1 - 2pp_2)^n) \\ \Pr[e_k = c] &= cp_e + (1-c)(1-p_e) \\ \Pr[Y_k = 1 \oplus b \oplus c] &= \frac{1 \oplus b \oplus c}{2}(1 - (1 - 2pp_1)^n) + \frac{b \oplus c}{2}(1 + (1 - 2pp_1)^n) \end{aligned}$$

Proof. Follows directly from the linearity of the expectation and Lemma 2 $\square$

Theorem 11 (Variance)

Let e' be the error vector, as described in Definition 5. Let further E denote the expectation of $\text{weight}(e')$. The variance of $\text{weight}(e')$ is given by

$$\text{Var}(\text{weight}(e')) = E - E^2 + 2 \sum_{0 \leq k < l < n} \sum_{b, b', c, c' \in \{0,1\}} \Pr[X_k = b, X_l = b'] \Pr[Y_k = c, Y_l = c'] \Pr[e_k = 1 \oplus b \oplus c, e_l = 1 \oplus b' \oplus c']$$

where

$$\begin{aligned} \Pr[X_k = b, X_l = b'] &= \sum_{r=0}^n \sum_{s=0}^r \sum_{d \in \{0,1\}} \text{pmf}_B(n, p_2, k, l, r, s) \Pr[\bigoplus_{j=1}^s x_{i_j} = d] \Pr[\bigoplus_{j=1}^{r-s} x_{k_j} = b \oplus d] \Pr[\bigoplus_{j=1}^{r-s} x_{l_j} = b' \oplus d] \\ \Pr[Y_k = c, Y_l = c'] &= \sum_{r=0}^n \sum_{s=0}^r \sum_{d \in \{0,1\}} \text{pmf}_B(n, p_1, k, l, r, s) \Pr[\bigoplus_{j=1}^s y_{i_j} = d] \Pr[\bigoplus_{j=1}^{r-s} y_{k_j} = c \oplus d] \Pr[\bigoplus_{j=1}^{r-s} y_{l_j} = c' \oplus d] \\ \Pr[e_k = 1 \oplus b \oplus c, e_l = 1 \oplus b' \oplus c'] &= [(1 \oplus b \oplus c)p_e + (b \oplus c)(1 - p_e)][(1 \oplus b' \oplus c')p_e + (b' \oplus c')(1 - p_e)] \end{aligned}$$

The probabilities $\text{pmf}_B(n, p_1, k, l, r, s)$ and $\text{pmf}_B(n, p_2, k, l, r, s)$ are computed according to Algorithm 4 over the sample spaces $B(n, p_1)$ and $B(n, p_2)$ respectively. The probabilities $\Pr[\bigoplus_{i=1}^N x_i = b]$ and $\Pr[\bigoplus_{i=1}^N y_i = b]$, for an arbitrary bit b and a nonnegative integer N , are computed according to Lemma 2 for $x, y \sim B(n, p)$. $\square$

6.3 Validation and Comparative Analysis

Experimental Validation. We experimentally validate our analytical formulae, derived earlier in this section for the variance of the weight of the error vector e' in the HQC scheme. In the uniform slice setting, the polynomials x, y, r_1, r_2 , and e are w -weighted polynomials of degrees less than $n = 250$, with $w \leq n/2$. Whereas in the binomial setting, the aforementioned quantities are independent polynomials of degrees less than $n = 250$, with Bernoulli(p) coefficients. The empirical variance of $\text{weight}(e')$ is computed over 100,000 trials. For each trial, we generate e' from the required distribution, record the weight $Z = \text{weight}(e')$ and its square Z^2 , and compute the empirical variance as $\langle Z^2 \rangle - \langle Z \rangle^2$, where $\langle \cdot \rangle$ is the sample mean. The close correspondence between the 'Theory' and 'Experiments' curves in Figure 8 demonstrates the correctness of our expressions.

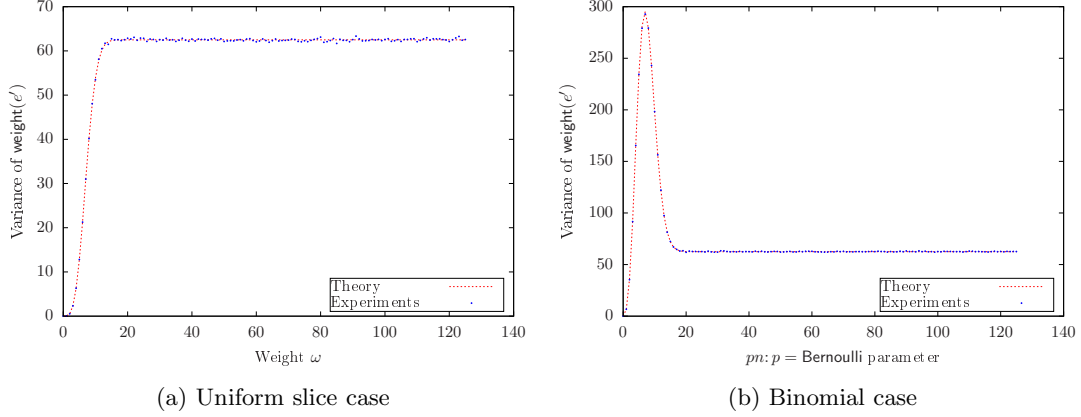


Fig. 8: Variance of the weight of the decryption vector e' in the HQC scheme

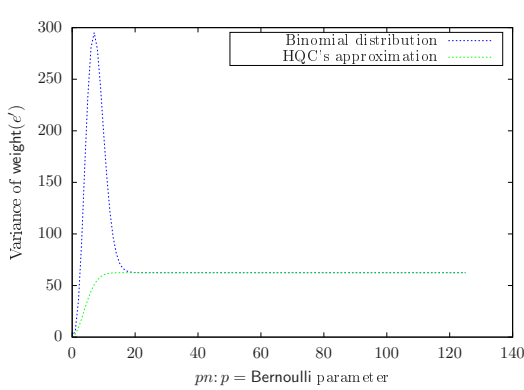


Fig. 9: Comparison of the variance of $\text{weight}(e')$ with the approximation from [3]

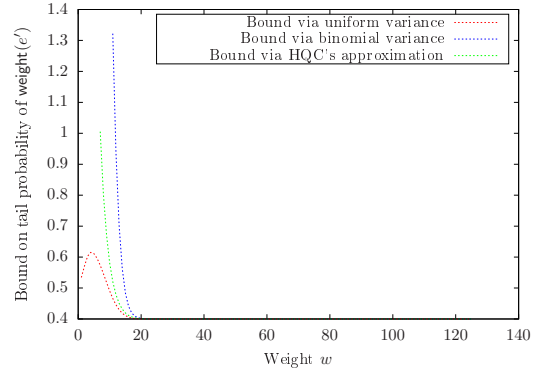


Fig. 10: Upper bound on the tail probability $\Pr[\text{weight}(e') \notin (1 \pm 0.1)\mathbb{E}(\text{weight}(e'))]$ for $n = 250$ and $w \in [0, 125]$

Comparison with Existing Approximations. Figure 9 compares the exact value of the variance of $\text{weight}(e')$ in the binomial setting, with the approximation from [3]. We observe again that this approximation is only accurate for relatively large values of p . We cannot employ the approximation from [4] because an upper bound on the variance of the weight of the polynomial product $x \cdot r_2$ (or $y \cdot r_1$) cannot be straightforwardly translated into an upper bound for the variance of $\text{weight}(e')$.

Tail Probability of $\text{weight}(e')$. We compute an upper bound, using Chebyshev’s inequality, on the probability that $\text{weight}(e') \notin (1 \pm 0.1)\mathbb{E}(\text{weight}(e'))$, for $n = 250$, across both the uniform slice and binomial models. In the uniform slice model, x, y, r_1, r_2, e are w -weighted polynomials, whereas in the binomial setting $x, y, r_1, r_2, e \sim B(n, \frac{w}{n})$. We also compare our computed bounds with the approximation from [3] (Figure 10). We remark again that $\text{weight}(e')$ is tightly concentrated around its mean in the uniform setting. This corroborates [3]’s heuristics of the scheme’s security under the uniform slice model. Moreover, the explicit formulae we derived for the expectation and variance of $\text{weight}(e')$ enable precise parameter tuning to further minimize tail probabilities and optimize the decryption failure rate of the scheme.

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7 Joint PMF of $(|S_k|, |S_k \cap S_l|)$ in the Binomial Model

In this section, we compute the joint PMF of the random variables $(|S_k|, |S_k \cap S_l|)$, for $S_j = \{(j - i) \bmod n : x_i = 1\}$, $j = k, l$, over the set $B(n, p)$ of n -bit vectors $(x_0, \dots, x_{n-1})$ with iid Bernoulli(p) coordinates.

As in the uniform slice case, we have:

$$\begin{aligned} e \in S_k \cap S_l &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } e = (k - i) \bmod n \text{ and } e = (l - j) \bmod n \\ &\iff \exists i, j \in \mathbb{Z}_n : x_i = x_j = 1 \text{ and } \{j - i = l - k \text{ or } i - j = n + k - l\} \end{aligned}$$

Therefore, $\Pr[|S_k| = r, |S_k \cap S_l| = s]$ corresponds to the probability that $\text{weight}(x) = r$, and there exist exactly s pairs (i, j) among the following pairs:

$$\{(0, l - k), \dots, (n + k - l - 1, n - 1)\} \cup \{(n + k - l, 0) \dots (n - 1, l - k - 1)\}$$

satisfying $x_i = x_j = 1$.

We need to distinguish the cases: (1) The case ' $n + k - l = l - k$ ', and (2) the case ' $n + k - l \neq l - k$ '.

7.1 The Case ' $n + k - l = l - k$ '

Theorem 12

Given a random n -bit vector $x = (x_0, \dots, x_{n-1}) \sim B(n, p)$. For nonnegative integers k and l satisfying $k < l$ and $l - k = n + k - l$, the joint probability mass function of $(|S_k|, |S_k \cap S_l|)$ is given by

$$\Pr[|S_k| = r, |S_k \cap S_l| = s] = \begin{cases} 0 & \text{if } [(s \bmod 2 = 1) \text{ or } (r > n) \text{ or } (s > r)] \\ \sum_{r_0=0}^r \binom{l-k}{s/2} \binom{l-k-s/2}{r_0-s/2} \binom{l-k-r_0}{r-r_0-s/2} p^r (1-p)^{n-r} & \text{otherwise} \end{cases} \quad (11)$$

Proof (Sketch). Note that $l - k = n/2$. Since the special cases are straightforward, we focus on the general case:

$$\begin{aligned} \Pr[|S_k| = r, |S_k \cap S_l| = s] &= \Pr[\text{weight}(x) = r, |S_k \cap S_l| = s] \\ &= \sum_{r_0=0}^r \Pr[\text{weight}(x_0, \dots, x_{l-k-1}) = r_0, \text{weight}(x) = r, |S_k \cap S_l| = s] \\ &= \sum_{r_0=0}^r \binom{l-k}{s/2} \binom{l-k-s/2}{r_0-s/2} p^{r_0} (1-p)^{n/2-r_0} \binom{l-k-r_0}{r-r_0-s/2} p^{r-r_0} (1-p)^{n/2-(r-r_0)} \\ &= \sum_{r_0=0}^r \binom{l-k}{s/2} \binom{l-k-s/2}{r_0-s/2} \binom{l-k-r_0}{r-r_0-s/2} p^r (1-p)^{n-r} \end{aligned}$$

The penultimate equation is justified by an argument analogous to that in Theorem 4's proof, with the following modification: the process of setting the coordinates for the subvector $(x_0, \dots, x_{n/2-1})$ now requires multiplication by $p^{r_0} (1-p)^{n/2-r_0}$, and likewise, setting the coordinates for $(x_{n/2}, \dots, x_{n-1})$ requires multiplication by $p^{r-r_0} (1-p)^{n/2-(r-r_0)}$. $\square$

7.2 The Case ' $n + k - l \neq l - k$ '

To compute the joint PMF of $(|S_k|, |S_k \cap S_l|)$ in case $n + k - l \neq l - k$, we proceed by analogy with the method established for the uniform case. In fact, we consider an undirected graph $\tilde{G}$ on the vertices $x_0, \dots, x_{n-1}$ that are iid Bernoulli random variables with parameter p , and where adjacency is defined by the rule: vertex x_i is adjacent to vertex x_j if and only if $(i - j) \bmod n \in \{l - k, n + k - l\}$.

We therefore reduce the probability that $(|S_k|, |S_k \cap S_l|) = (r, s)$ to the probability that $\text{weight}(x_0, \dots, x_{n-1}) = r$ and there exist exactly s edges in $\tilde{G}$ that are incident to vertices both equal to 1.

Likewise, the analysis established for the shape of graph $\tilde{G}$ in the uniform case holds verbatim here. Thus, we omit a detailed repetition and proceed immediately to compute the aforementioned probability when: (1) $\tilde{G}$ is a cycle graph, and (2) $\tilde{G}$ is a disjoint union of cycle graphs.

7.2.1 Cycle Graphs

Let i, j, s be nonnegative integers less than or equal to n . Let further p be a real number in the closed interval $[0, 1]$.

Definition 6 (Cycle-PMF in the Binomial Model). For $i > 0$, consider an i -bit vector $(x_0, \dots, x_{i-1}) \in \{0, n\}^i$ with iid Bernoulli(p) coordinates.

$\text{cycle-pmf}_b(i, j, s)$ denotes the probability that the cycle graph $x_0 - \dots - x_{i-1} - x_0$ has exactly j vertices equal to 1 and s edges incident to vertices both equal to 1. The probability is taken over all $(x_0, \dots, x_{i-1}) \sim B(i, p)$.

Remark 1. Note that $\text{cycle-pmf}_b(i, j, s)$ and $\text{cycle-pmf}(i, j, s)$ are defined over different probability spaces. For $\text{cycle-pmf}_b(i, j, s)$, the space consists of all i -bit vectors where each coordinate is set independently with probability p . In contrast, for $\text{cycle-pmf}(i, j, s)$, the space consists of all i -bit vectors with *exactly* j bits set. In the latter case, the weight j is a fixed parameter of the space itself, not a random outcome.

We introduce now the auxiliary functions needed to compute $\text{cycle-pmf}_b(i, j, s)$.

Definition 7 (Functions $\tilde{f}_0, \tilde{f}_1, \tilde{g}, \tilde{f}$). Let i, j, s be nonnegative integers, and p a real number in $[0, 1]$. Let further $x_0 - \dots - x_i$ be a linear chain, where $(x_0, \dots, x_i)$ is an $(i + 1)$ -bit random vector with iid Bernoulli(p) coordinates.

$\tilde{f}_0(i, j, s)$ denotes the conditional probability that the linear chain $x_0 - \dots - x_i$ has weight j and contains exactly s edges with both endpoints equal to 1, given that x_0 is equal to 0.

$\tilde{f}_1(i, j, s)$ denotes the conditional probability that the linear chain $x_0 - \dots - x_i$ has weight j and contains exactly s edges with both endpoints equal to 1, given that x_0 is equal to 1.

$\tilde{g}(i, j, s, b, c)$, for $b, c \in \{0, 1\}$, denotes the conditional probability that the linear chain $x_0 - \dots - x_i$ has weight j and contains exactly s edges with both endpoints equal to 1, given that x_0 and x_i are equal to b and c , respectively.

$\tilde{f}(i, j, s)$ denotes the probability that the linear chain $x_0 - \dots - x_i$ has weight j and contains exactly s edges with both endpoints equal to 1.

The probability is taken over all $(i + 1)$ -bit vectors $(x_0, \dots, x_i)$, where each coordinate is set independently with parameter p .

Remark 2. Note again the difference between the uniform slice and binomial cases in the definition of the auxiliary functions $\tilde{f}_0, \tilde{f}_1, \tilde{g}, \tilde{f}$: the probability is taken, in the binomial case, over $(i + 1)$ -bit vectors $(x_0, \dots, x_i)$, where each coordinate is set independently with parameter p . Thus, the weight j is a random outcome and not a fixed parameter of the probability space, unlike in the uniform slice case.

Let us proceed now to the computation of the above functions. Again, we employ a dynamic programming approach to compute $\tilde{f}_0, \tilde{f}_1$, and $\tilde{g}$.

Form. 6. and **Form. 7.** give respectively the formulae for computing $\tilde{f}_0, \tilde{f}_1$, and $\tilde{g}$; the base cases are immediate, whilst the recurrence relations follow from

$$\begin{aligned}\tilde{f}_0(i, j, s) &= \Pr[x_1 = 0 \mid x_0 = 0] \tilde{f}_0(i - 1, j, s) + \Pr[x_1 = 1 \mid x_0 = 0] \tilde{f}_1(i - 1, j, s) \\ \tilde{f}_1(i, j, s) &= \Pr[x_1 = 0 \mid x_0 = 1] \tilde{f}_0(i - 1, j - 1, s) + \Pr[x_1 = 1 \mid x_0 = 1] \tilde{f}_1(i - 1, j - 1, s - 1) \\ \tilde{g}(i, j, s, b, c) &= \Pr[x_1 = 0 \mid x_0 = b, x_i = c] \tilde{g}(i - 1, j - b, s, 0, c) + \Pr[x_1 = 1 \mid x_0 = b, x_i = c] \\ &\quad \tilde{g}(i - 1, j - b, s - b, 1, c)\end{aligned}$$

Where

$$\begin{aligned}\Pr[x_1 = 0 \mid x_0 = 0] &= \Pr[x_1 = 0 \mid x_0 = 1] = \Pr[x_1 = 0] = 1 - p \\ \Pr[x_1 = 1 \mid x_0 = 0] &= \Pr[x_1 = 1 \mid x_0 = 1] = \Pr[x_1 = 1] = p \\ \Pr[x_1 = 0 \mid x_0 = b, x_i = c] &= 1 - p \text{ and } \Pr[x_1 = 1 \mid x_0 = b, x_i = c] = p\end{aligned}$$

$\tilde{f}_0(i, j, s) = \tilde{f}_1(i, j, s) = 0$ if $(j > i + 1) \vee (s > i) \vee (s > j - 1) \vee (s < 0)$. Otherwise:	
$\tilde{f}_0(i, j, s) =$	$\begin{cases} (j = 0) \cdot (s = 0) & \text{if } i = 0 \\ (1 - p)(s = 0)(j = 0) + p(s = 0)(j = 1) & \text{if } i = 1 \\ \begin{cases} 0 & \text{if } j > i \\ (1 - p)\tilde{f}_0(i - 1, j, s) + p\tilde{f}_1(i - 1, j, s) & \text{otherwise} \end{cases} & \text{otherwise} \end{cases}$
$\tilde{f}_1(i, j, s) =$	$\begin{cases} (j = 1) \cdot (s = 0) & \text{if } i = 0 \\ (1 - p)(s = 0)(j = 1) + p(s = 1)(j = 2) & \text{if } i = 1 \\ \begin{cases} 0 & \text{if } j = 0 \text{ or } j - 1 > i \\ (1 - p)\tilde{f}_0(i - 1, j - 1, s) + p\tilde{f}_1(i - 1, j - 1, s - 1) & \text{otherwise} \end{cases} & \text{otherwise} \end{cases}$

Form. 6. Computation of $\tilde{f}_0$ and $\tilde{f}_1$

$\tilde{g}(i, j, s, b, c) = 0$ if $(j > i + 1) \vee (s > i) \vee (j > b + c) \vee (s < 0) \vee (i == 0)$. Otherwise:	
$\tilde{g}(i, j, s, b, c) =$	$\begin{cases} (j = b + c) \cdot (s = b \cdot c) & \text{if } i = 1 \\ (1 - p)\tilde{g}(i - 1, j - b, s, 0, c) + p\tilde{g}(i - 1, j - b, s - b, 1, c) & \text{otherwise} \end{cases}$

Form. 7. Computation of $\tilde{g}$

$\tilde{f}(i, j, s) = (1 - p)\tilde{f}_0(i, j, s) + p\tilde{f}_1(i, j, s)$
--

Form. 8. Computation of $\tilde{f}$

$\text{cycle-pmf}_b(i, j, s) =$	$\begin{cases} 0 & \text{if } (i = 0) \vee (j > i) \vee (s > i) \vee (s > j) \\ \tilde{f}(i - 1, j, s) & \text{if } (i < 3), (i > 0) \\ p^i(s = i)(j = i) & \text{if } (s = i) \vee (j = i) \\ \left[(1 - p)\tilde{f}(p, i - 2, j, s) + \right. \\ \left. p \sum_{b, c \in \{0, 1\}} [(pb + (1 - p)(1 - b))(pc + (1 - p)(1 - c)) \right. \\ \left. \tilde{g}(i - 2, j - 1, s - b - c, b, c)] \right] & \text{otherwise} \end{cases}$
---------------------------------	---

Form. 9. Computation of cycle-pmf_b

Remark 3. Following the same approach outlined in the uniform case, we precompute $\tilde{f}_0(i, j, s)$, $\tilde{f}_1(i, j, s)$, and $\tilde{g}(i, j, s, b, c)$, for all nonnegative integers $0 \leq i, j, s \leq n$ and all bits b, c , via bottom-up dynamic programming. The $O(n^3)$ precomputation cost in both time and space allows for a constant time computation of cycle-pmf_b .

7.2.2 Disjoint Union of Cycle Graphs

Suppose now that $\tilde{G}$ is disjoint union of d cycle graphs $\tilde{G} = \bigcup_{i=0 \dots d-1} \tilde{G}_i$, where $d > 1$. Recall that $\tilde{G}$ is on vertices $x_0, \dots, x_{n-1}$ that are iid Bernoulli(p).

We maintain the notation from the uniform case: S denotes the total number of egdes in $\tilde{G}$ that are incident to two vertices both assigned the value 1. Moreover, $\text{weight}(\tilde{G})$, which is now a random variable that we denote R , refers to the weight of $\tilde{G}$ or equivalently the total number of vertices that are equal to 1. Correspondingly, for $0 \leq i \leq d - 1$, S_i denotes the total number

of edges in $\tilde{G}_i$ that are incident to two vertices both assigned the value 1, and $\text{weight}(\tilde{G}_i)$ refers to the number of vertices that are equal to 1 in $\tilde{G}_i$.

$$\begin{aligned} \Pr[R = r, S = s] &= \sum_{\substack{r_0, \dots, r_{d-1} \\ \sum r_i = r}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \Pr\left[\bigwedge_{i=0}^{d-1} (\text{weight}(\tilde{G}_i) = r_i, S_i = s_i)\right] \\ &= \sum_{\substack{r_0, \dots, r_{d-1} \\ \sum r_i = r}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \prod_{i=0}^{d-1} \text{cycle-pmf}_b(p, n/d, r_i, s_i) \end{aligned}$$

Likewise, $\Pr[R = r, S = s]$ can be computed in a dynamic programming approach. Let D be the largest divisor of n , and consider the following function:

$$\begin{aligned} \tilde{h}: [0, D] \times [0, n] \times [0, n] \times [0, D] &\longrightarrow \mathbb{R} \\ (N, r, s, d) &\longmapsto \sum_{\substack{r_0, \dots, r_{d-1} \\ \sum r_i = r}} \sum_{\substack{s_0, \dots, s_{d-1} \\ \sum s_i = s}} \prod_{i=0}^{d-1} \text{cycle-pmf}_b(N, r_i, s_i) \end{aligned}$$

Then, $\tilde{h}$ can be computed recursively using the following recurrence:

$$\begin{aligned} \tilde{h}(N, r, s, d) &= \sum_{r_0=0}^r \sum_{s_0=0}^s \text{cycle-pmf}_b(N, r_0, s_0) \sum_{\substack{r_1, \dots, r_{d-1} \\ \sum r_i = r - r_0}} \sum_{\substack{s_1, \dots, s_{d-1} \\ \sum s_i = s - s_0}} \prod_{i=1}^{d-1} \text{cycle-pmf}_b(N, r_i, s_i) \\ &= \sum_{w_0=0}^w \sum_{s_0=0}^s \text{cycle-pmf}_b(N, r_0, s_0) \tilde{h}(N, r - r_0, s - s_0, d - 1) \end{aligned} \tag{12}$$

The algorithm for computing the joint PMF of $(|S_k|, |S_k \cap S_l|)$ is straightforward.

Algorithm 4: Computation of the joint PMF of $(|S_k|, |S_k \cap S_l|)$

Input: nonnegative integers n, k, l, r, s with $k < l < n$ and $s, r \leq n$, and a real number $p \in [0, 1]$

Output: $\Pr[|S_k| = r, |S_k \cap S_l| = s] = \text{pmf}_B(n, p, k, l, r, s)$

```

1 if  $n + k - l = l - k$  then
     $res = \begin{cases} 0 & \text{if } [(s \bmod 2 = 1) \text{ or } \\ & (r > n) \text{ or } (s > r)] \\ \sum_{r_0=0}^r \binom{l-k}{s/2} \binom{l-k-s/2}{r_0-s/2} \binom{l-k-r_0}{r-r_0-s/2} p^r (1-p)^{n-r} & \text{otherwise} \end{cases}$ 
2 else
    // We assume the values  $\tilde{f}_0(i, j, t)$ ,  $\tilde{f}_1(i, j, t)$ ,  $\tilde{g}(i, j, t, b, c)$ , and
    //  $\tilde{h}(N, j, t, f)$ , for all parameter sets under consideration, are
    // precomputed using dynamic programming as mentioned earlier
3  $d = \gcd(l - k, n)$ ;
4 if  $d = 1$  then
5      $res = \text{cycle-pmf}_b(n, r, s)$ ;      /* cycle-pmfb is computed according to Form. 9. */
6 else
7      $res = \tilde{h}(n/d, r, s, d)$ ;
8 return  $res$ ;

```

Remark 4. We remark that, besides depending on n and p , the joint PMF of $(|S_k|, |S_k \cap S_l|)$ is a function solely of $\gcd(l-k, n)$. Consequently, the same optimization, detailed in Subsubsection 5.3.2, remains applicable here for the binomial case.

7.3 Experimental Validation

Similarly to the uniform slice case, we validate experimentally our theoretical formulae for the joint probability $\Pr[|S_k| = r, |S_k \cap S_l| = s]$. We do this using a sample size of 100,000. In each sample, we generate a polynomial of degree less than $n = 250$ and with Bernoulli($p = 0.26$) coefficients, then record the values $R = |S_k|$ and $S = |S_k \cap S_l|$; the empirical value of $\Pr[|S_k| = r, |S_k \cap S_l| = s]$ is estimated as the proportion of trials where $R = r$ and $S = s$. We plot in Figure 11 this joint PMF for the three values: (i) $l-k = 17$, corresponding to the case ' $\gcd(l-k, n) = 1$ ', (ii) $l-k = 25$, corresponding to the case ' $\gcd(l-k, n) \neq 1$ ', and (iii) $l-k = 125$ corresponding to the case ' $l-k = n + k - l$ '. The strong correspondance between the 'Theory' and 'Experiments' curves verifies our formulae.

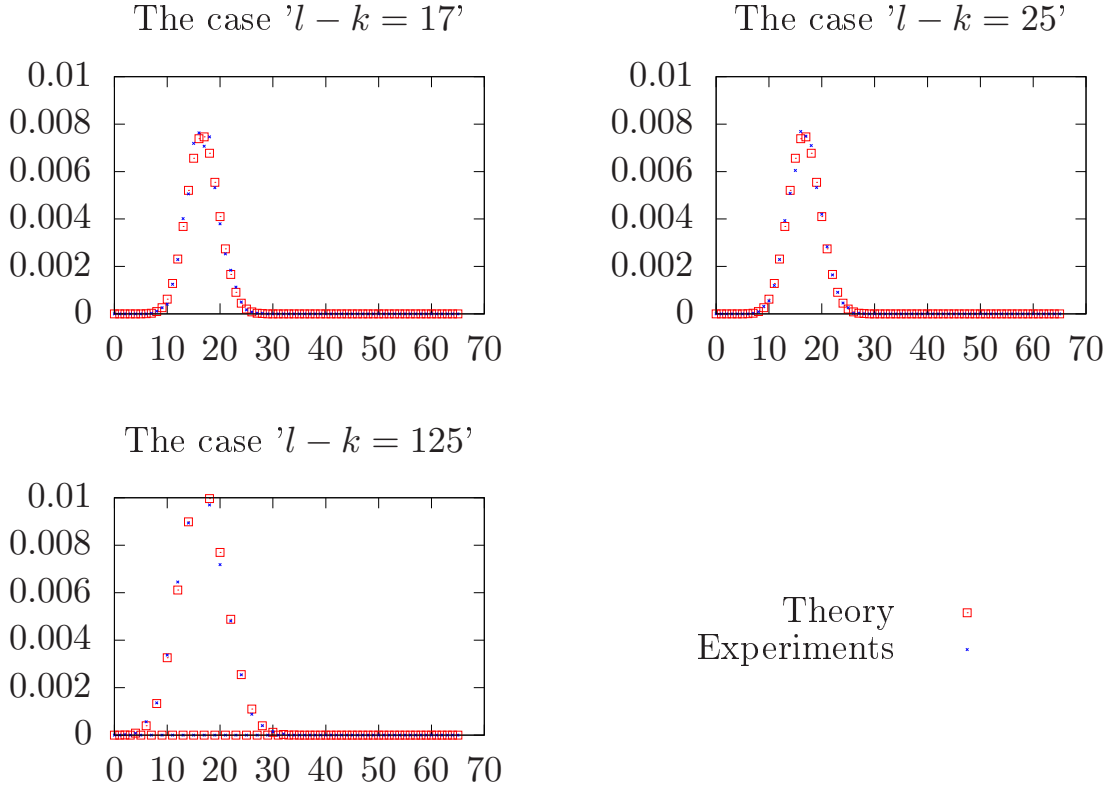


Fig. 11: Joint probability $\Pr[|S_k| = r, |S_k \cap S_l| = s]$ for $n = 250$, $p = 0.26$, and $r = 65$. The x -axis corresponds to the values $s \in [0, r]$